



US012400507B2

(12) **United States Patent**
Bodurka et al.

(10) **Patent No.:** **US 12,400,507 B2**
(45) **Date of Patent:** ***Aug. 26, 2025**

(54) **ENTRY ACCESS SYSTEM AND METHOD
FOR QUESTIONNAIRE SCREENING**

(71) Applicant: **Masonite Corporation**, Tampa, FL
(US)

(72) Inventors: **Alex Bodurka**, Portage, MI (US); **Jan
H. Ettrich**, Geneva, IL (US); **David
Toll**, Cicero, IN (US)

(73) Assignee: **Masonite Corporation**, Tampa, FL
(US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

This patent is subject to a terminal dis-
claimer.

(21) Appl. No.: **18/342,326**

(22) Filed: **Jun. 27, 2023**

(65) **Prior Publication Data**

US 2023/0419759 A1 Dec. 28, 2023

Related U.S. Application Data

(63) Continuation of application No. 17/473,492, filed on
Sep. 13, 2021, now Pat. No. 11,727,740.

(60) Provisional application No. 63/077,193, filed on Sep.
11, 2020.

(51) **Int. Cl.**
G07C 9/27 (2020.01)
G07C 9/10 (2020.01)
G07C 9/25 (2020.01)
G16H 10/20 (2018.01)

(52) **U.S. Cl.**
CPC **G07C 9/27** (2020.01); **G07C 9/10**
(2020.01); **G07C 9/25** (2020.01); **G16H 10/20**
(2018.01)

(58) **Field of Classification Search**

CPC **G07C 9/27**; **G07C 9/10**; **G07C 9/25**
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

7,824,200 B2 11/2010 Bryla et al.
8,156,671 B2 4/2012 Presley et al.
8,169,169 B2 5/2012 Hass et al.

(Continued)

FOREIGN PATENT DOCUMENTS

AU 2020101466 A4 8/2020
CN 200968110 Y 10/2007

(Continued)

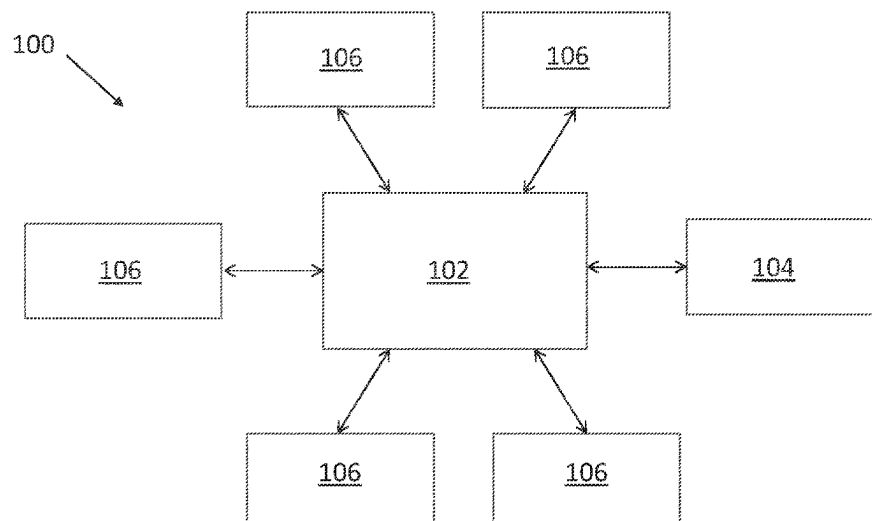
Primary Examiner — Nabil H Syed

(74) *Attorney, Agent, or Firm* — IceMiller LLP

(57) **ABSTRACT**

An entry system includes a plurality of first user devices, such as key fobs, ID cards and the like. A plurality of second user devices are maintained by users, and may include mobile phones, smart watches and tablets. A main entry system controls a barrier to allow entry to approved individuals. An authorization system is configured to validate the first and second user devices, and to send an electronic questionnaire to the second user device. The user must complete the questionnaire and transmit it to the main entry system. The second user device, the main entry system, and the authorization system communicate with each other over a computer network. If the user successfully completes the questionnaire and the user's identity is authenticated, the main entry system allows the barrier to be opened so that the user may enter the facility.

20 Claims, 29 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

8,337,039 B1 12/2012 Larkin
 8,354,914 B2 1/2013 Buckingham et al.
 8,772,970 B2 7/2014 Lambrou
 9,290,966 B2 3/2016 Hanchett, Jr.
 9,652,917 B2 5/2017 Johnson et al.
 9,704,320 B2 7/2017 Johnson et al.
 9,922,513 B1 3/2018 Hall et al.
 10,037,636 B2 7/2018 Ho et al.
 10,043,332 B2 8/2018 Scalisi et al.
 10,361,880 B1 7/2019 Marcinkowski et al.
 10,453,280 B2 10/2019 Kontturi
 10,664,688 B2 5/2020 Goulden et al.
 10,691,953 B2 6/2020 Johnson et al.
 10,733,823 B2 8/2020 Scalisi
 10,755,509 B2 8/2020 Saeedi et al.
 10,803,685 B2 10/2020 Marcinkowski et al.
 10,818,118 B2 10/2020 Sivalingam et al.
 10,832,509 B1 11/2020 Read et al.
 10,938,250 B2 3/2021 Ahmed et al.
 10,950,076 B1 3/2021 Hall et al.
 10,964,138 B2 3/2021 Geerlings et al.
 10,997,547 B2 5/2021 Hall et al.
 10,999,092 B2 5/2021 Marcinkowski et al.
 11,174,666 B2 11/2021 Demele et al.
 11,244,523 B1 2/2022 Brady et al.
 11,248,410 B2 2/2022 Dreyer et al.
 11,290,298 B2 3/2022 Marcinkowski et al.
 11,339,604 B2 5/2022 Dreyer
 11,346,141 B2 5/2022 Gregoriou
 11,346,150 B1 5/2022 Johnston
 11,352,812 B2 6/2022 Johnson
 11,373,471 B2 6/2022 Anderson et al.
 D957,688 S 7/2022 Kim et al.
 11,388,373 B2 7/2022 Scalisi
 11,403,899 B2 8/2022 Zhang
 11,403,902 B2 8/2022 Ho et al.
 11,417,200 B2 8/2022 Hass et al.
 11,441,332 B2 9/2022 Johnson et al.
 11,532,192 B2 12/2022 Hass et al.
 11,536,078 B2 12/2022 Dreyer
 11,539,794 B1 12/2022 Weber et al.
 11,543,801 B2 1/2023 Hall et al.
 11,655,669 B2 5/2023 Hutton et al.
 11,694,287 B2 7/2023 Marcinkowski et al.
 11,729,014 B2 8/2023 Marcinkowski et al.
 11,767,710 B1 9/2023 Johnston
 12,141,130 B2 11/2024 Gong et al.
 2014/0020295 A1 1/2014 Bonahoom et al.
 2014/0267739 A1 9/2014 Ibsies
 2016/0163139 A1 6/2016 Kankkunen et al.
 2016/0322847 A1 11/2016 Geiszler
 2017/0040827 A1 2/2017 Weber
 2017/0358952 A1 12/2017 Butler et al.
 2019/0271186 A1 9/2019 Chen et al.
 2019/0333302 A1 10/2019 Kagnev et al.
 2019/0357050 A1* 11/2019 Kennedy-Foster
 H04L 63/0853
 2020/0014552 A1 1/2020 Tan et al.
 2020/0082240 A1 3/2020 Heitmar
 2020/0322917 A1* 10/2020 Rosenberg H04W 64/003
 2020/0349786 A1 11/2020 Ho et al.
 2020/0378172 A1 12/2020 Lerpard
 2021/0207419 A1 7/2021 Sorice et al.
 2021/0207420 A1 7/2021 Sorice et al.
 2021/0207421 A1 7/2021 Sorice et al.
 2021/0209878 A1 7/2021 Kim et al.
 2021/0375084 A1* 12/2021 Aubrey G07C 9/27
 2022/0068070 A1 3/2022 Ouellette et al.
 2022/0155005 A1 5/2022 Lee et al.
 2022/0186543 A1 6/2022 Soderqvist
 2022/0263297 A1 8/2022 Bodurka et al.
 2022/0351722 A1 11/2022 Mandry et al.
 2022/0392287 A1 12/2022 Shen

2023/0062621 A1 3/2023 Cheung
 2023/0184022 A1 6/2023 Eliasson et al.
 2024/0146046 A1 5/2024 Gong et al.

FOREIGN PATENT DOCUMENTS

CN 103606213 A 2/2014
 CN 104105248 A 10/2014
 CN 203925102 U 11/2014
 CN 205117039 U 3/2016
 CN 106761274 A 5/2017
 CN 106836990 A 6/2017
 CN 206309153 U 7/2017
 CN 206329255 U 7/2017
 CN 107191110 A 9/2017
 CN 107448128 A 12/2017
 CN 107575137 A 1/2018
 CN 107610390 A 1/2018
 CN 107643706 A 1/2018
 CN 206903524 U 1/2018
 CN 107705511 A 2/2018
 CN 107730687 A 2/2018
 CN 107747457 A 3/2018
 CN 107795255 A 3/2018
 CN 107829659 A 3/2018
 CN 108118987 A 6/2018
 CN 207458156 U 6/2018
 CN 108389290 A 8/2018
 CN 108399674 A 8/2018
 CN 108412359 A 8/2018
 CN 108442852 A 8/2018
 CN 108661486 A 10/2018
 CN 108756617 A 11/2018
 CN 108868508 A 11/2018
 CN 108915498 A 11/2018
 CN 108924015 A 11/2018
 CN 108952449 A 12/2018
 CN 109098543 A 12/2018
 CN 109138774 A 1/2019
 CN 109191617 A 1/2019
 CN 109191739 A 1/2019
 CN 109236134 A 1/2019
 CN 109236135 A 1/2019
 CN 109267837 A 1/2019
 CN 109472902 A 3/2019
 CN 208777869 U 4/2019
 CN 110009782 A 7/2019
 CN 110264682 A 9/2019
 CN 209429913 U 9/2019
 CN 110409954 A 11/2019
 CN 110505182 A 11/2019
 CN 110531631 A 12/2019
 CN 209990389 U 1/2020
 CN 110778265 A 2/2020
 CN 111173404 A 5/2020
 CN 111270959 A 6/2020
 CN 111343432 A 6/2020
 CN 111441680 A 7/2020
 CN 211038390 U 7/2020
 CN 211124167 U 7/2020
 CN 111505947 A 8/2020
 CN 111540093 A 8/2020
 CN 111653025 A 9/2020
 CN 111764747 A 10/2020
 CN 211851423 U 11/2020
 CN 112096221 A 12/2020
 CN 112211496 A 1/2021
 CN 112267796 A 1/2021
 CN 112330869 A 2/2021
 CN 112491668 A 3/2021
 CN 112539020 A 3/2021
 CN 213205374 U 5/2021
 CN 112901047 A 6/2021
 CN 113048374 A 6/2021
 CN 113129476 A 7/2021
 CN 113338772 A 9/2021
 CN 113345130 A 9/2021
 CN 113545639 A 10/2021
 CN 113645446 A 11/2021

(56)

References Cited

FOREIGN PATENT DOCUMENTS

CN 113674454 A 11/2021
 CN 113706744 A 11/2021
 CN 214835862 U 11/2021
 CN 214886488 U 11/2021
 CN 214943492 U 11/2021
 CN 215169563 U 12/2021
 CN 113947834 A 1/2022
 CN 217240775 U 8/2022
 CN 217307315 U 8/2022
 CN 115095246 A 9/2022
 CN 115164381 A 10/2022
 CN 115235096 A 10/2022
 CN 115294675 A 11/2022
 CN 115324428 A 11/2022
 CN 115341821 A 11/2022
 CN 115393988 A 11/2022
 CN 218454661 U 2/2023
 CN 115798084 A 3/2023
 CN 116251221 A 6/2023
 CN 219197192 U 6/2023
 CN 219246122 U 6/2023
 CN 116498183 A 7/2023
 CN 116760602 A 9/2023
 CN 117238059 A 12/2023
 DE 202022101334 U1 5/2022
 EP 3118405 A1 1/2017
 EP 3537398 A1 9/2019
 EP 3845729 A1 7/2021
 FR 3132160 A3 7/2023
 IN 201921017138 A 5/2019
 IN 201911049967 A 12/2019
 IN 201911048995 A 5/2021
 IN 202111053824 A 12/2021
 IN 202241020116 A 4/2022
 IN 202241026843 A 5/2022
 IN 411059 B 11/2022
 IN 202111013824 A 12/2022
 IN 430120 B 4/2023
 IN 202341019331 A 4/2023
 JP 6680821 B2 3/2020
 JP 2020042440 A 3/2020
 JP 6777955 B2 10/2020
 KR 20160025226 A 3/2016
 KR 20160050965 A 5/2016
 KR 20160123639 A 10/2016
 KR 20160124481 A 10/2016
 KR 101800514 B1 12/2017

KR 20180076537 A 7/2018
 KR 20180131716 A 12/2018
 KR 101988851 B1 6/2019
 KR 20190098719 A 8/2019
 KR 20190098720 A 8/2019
 KR 102024754 B1 9/2019
 KR 102079532 B1 2/2020
 KR 102150642 B1 9/2020
 KR 20200143302 A 12/2020
 KR 20210004253 A 1/2021
 KR 20210016726 A 2/2021
 KR 20210019211 A 2/2021
 KR 20210019216 A 2/2021
 KR 20210019218 A 2/2021
 KR 20210019241 A 2/2021
 KR 20210045195 A 4/2021
 KR 20210072510 A 6/2021
 KR 20210078199 A 6/2021
 KR 102303254 B1 9/2021
 KR 102329035 B1 11/2021
 KR 102341883 B1 12/2021
 KR 20210153308 A 12/2021
 KR 20220031428 A 3/2022
 KR 102421708 B1 7/2022
 KR 20220113136 A 8/2022
 KR 20220119878 A 8/2022
 KR 102472384 B1 12/2022
 KR 102534826 B1 5/2023
 TW M627626 U 6/2022
 WO 2013163124 A1 10/2013
 WO 2016032464 A1 3/2016
 WO 2016175910 A1 11/2016
 WO 2017024088 A1 2/2017
 WO 2018184450 A1 10/2018
 WO 2018184452 A1 10/2018
 WO 2018184453 A1 10/2018
 WO 2018184454 A1 10/2018
 WO 2019068021 A1 4/2019
 WO 2019162435 A1 8/2019
 WO 2022004989 A1 1/2022
 WO 2022010071 A1 1/2022
 WO 2022010072 A1 1/2022
 WO 2022010073 A1 1/2022
 WO 2022010074 A1 1/2022
 WO 2022020893 A1 2/2022
 WO 2022145966 A1 7/2022
 WO 2022263147 A1 12/2022
 WO 2023180427 A1 9/2023

* cited by examiner

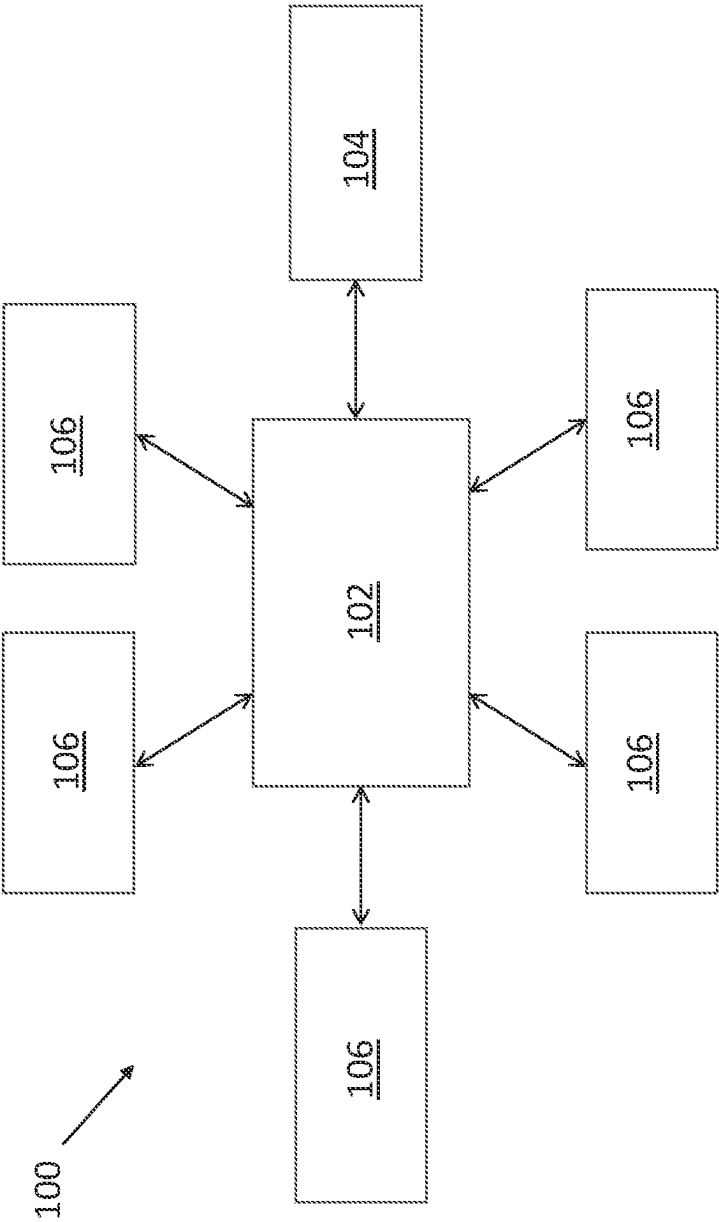


FIG. 1

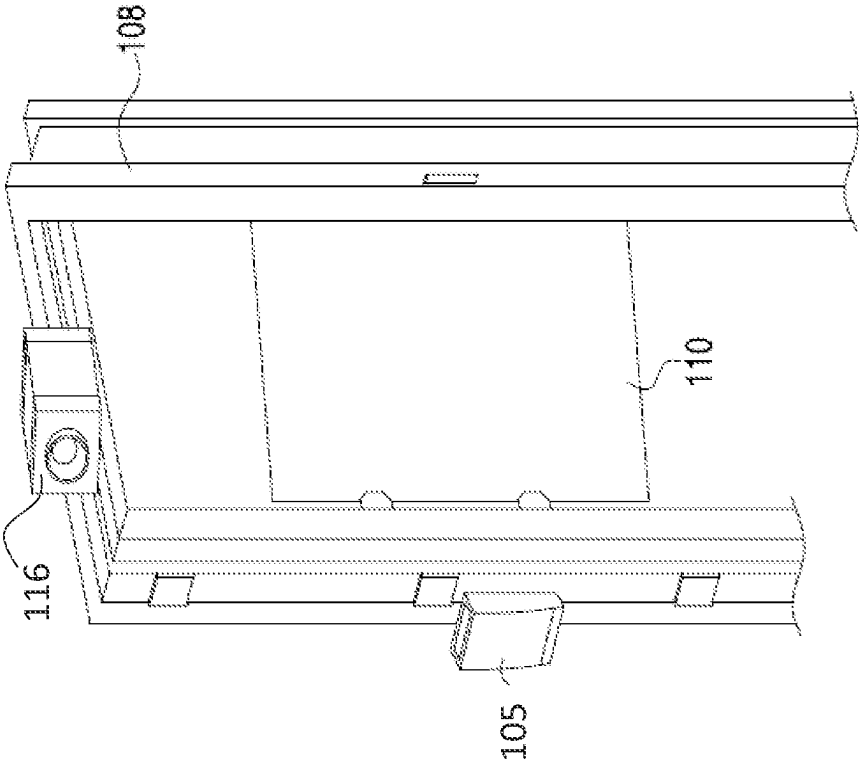


FIG. 2

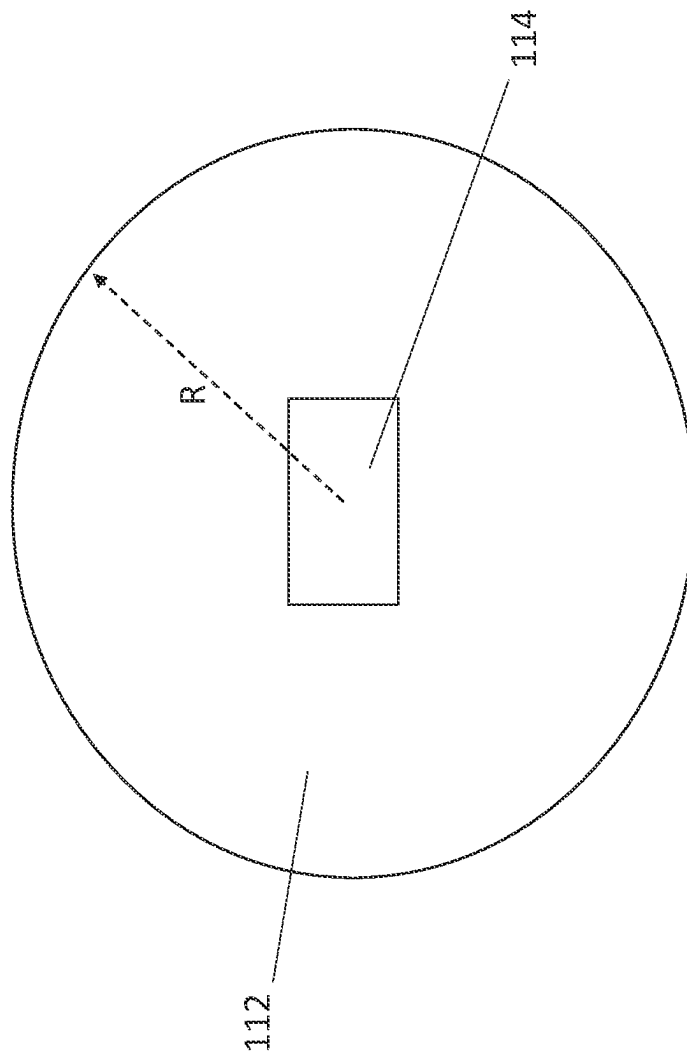


FIG. 3

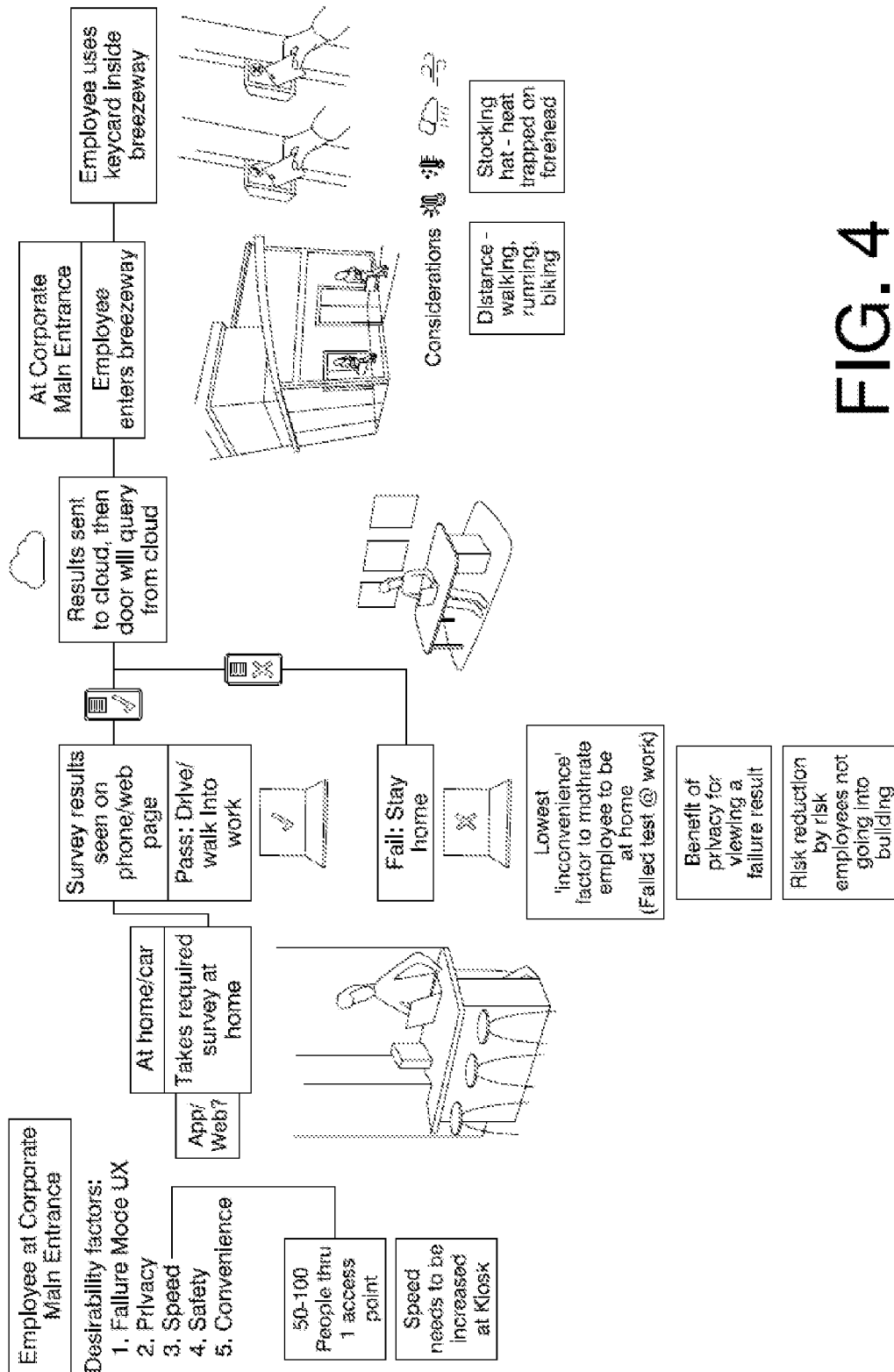


FIG. 4

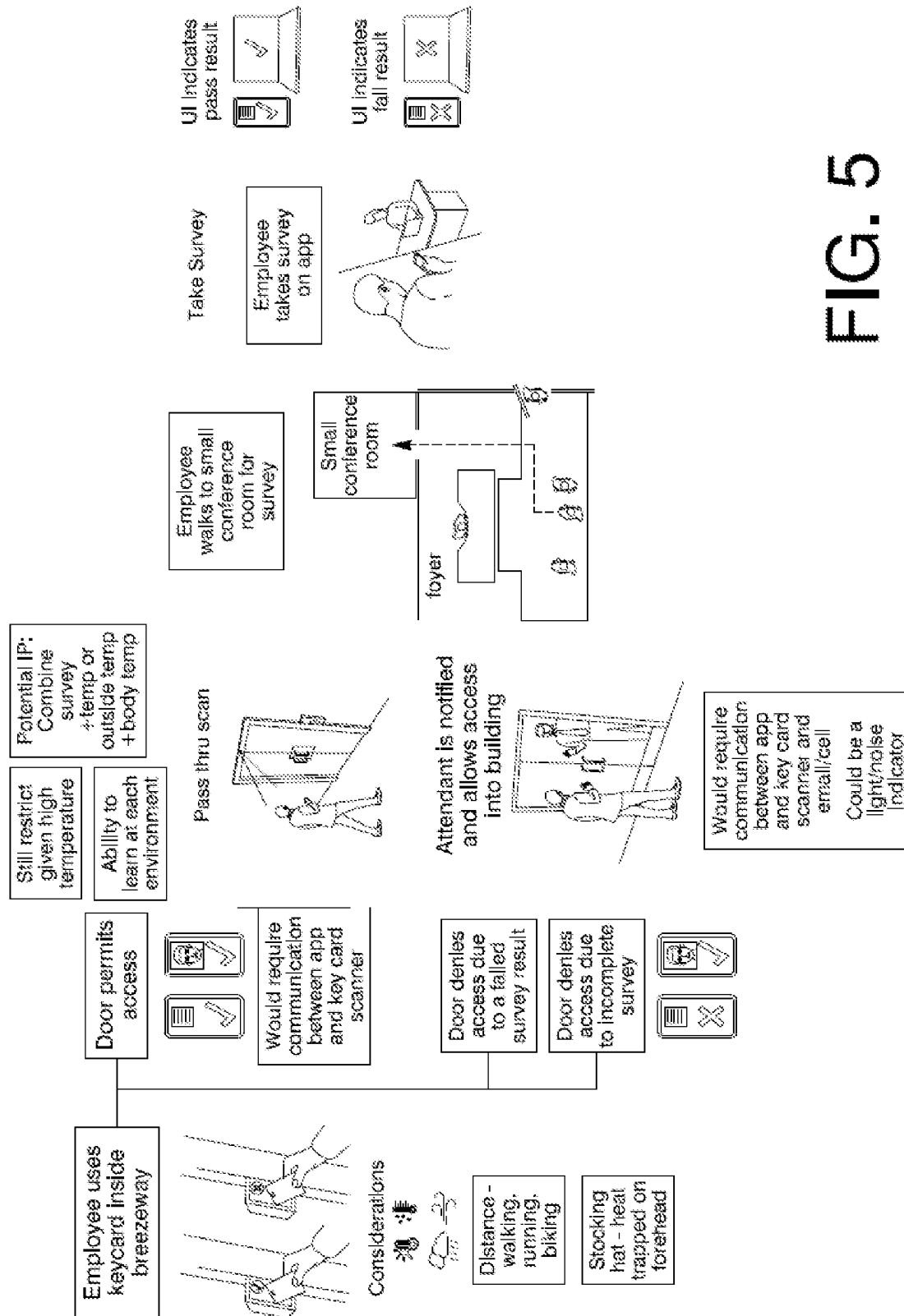


FIG. 5

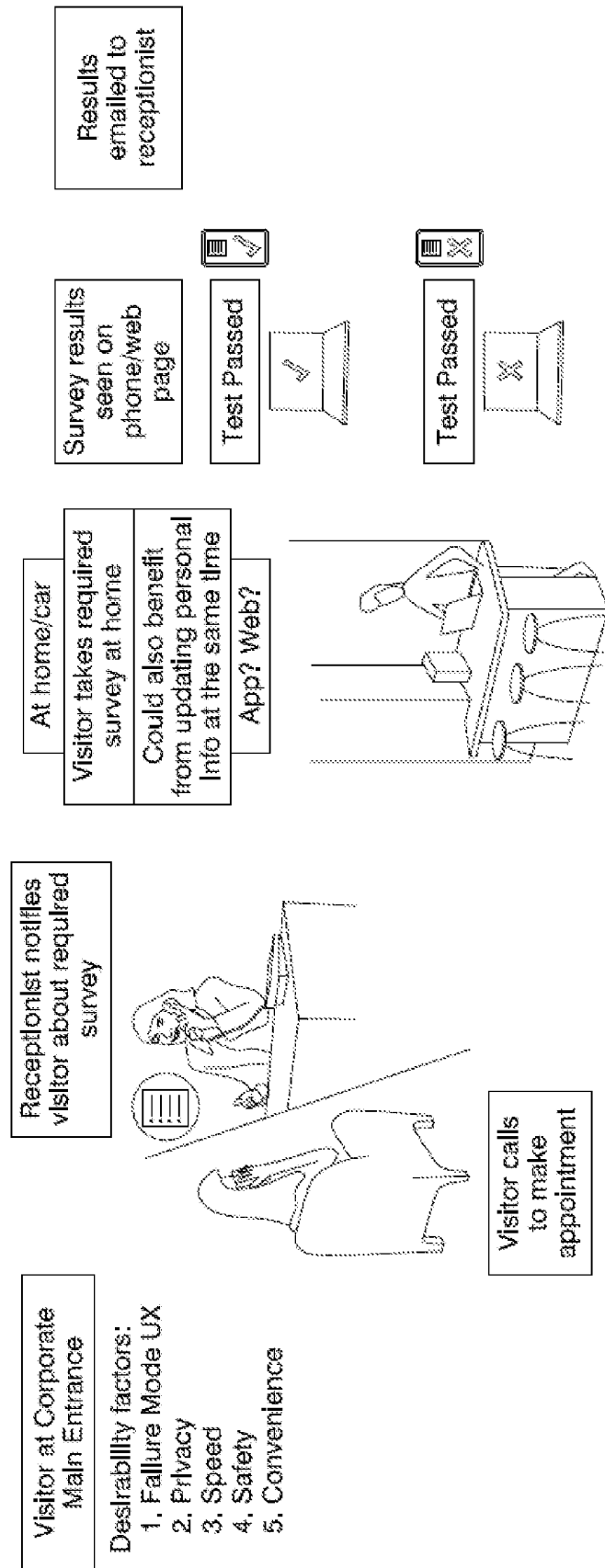


FIG. 6

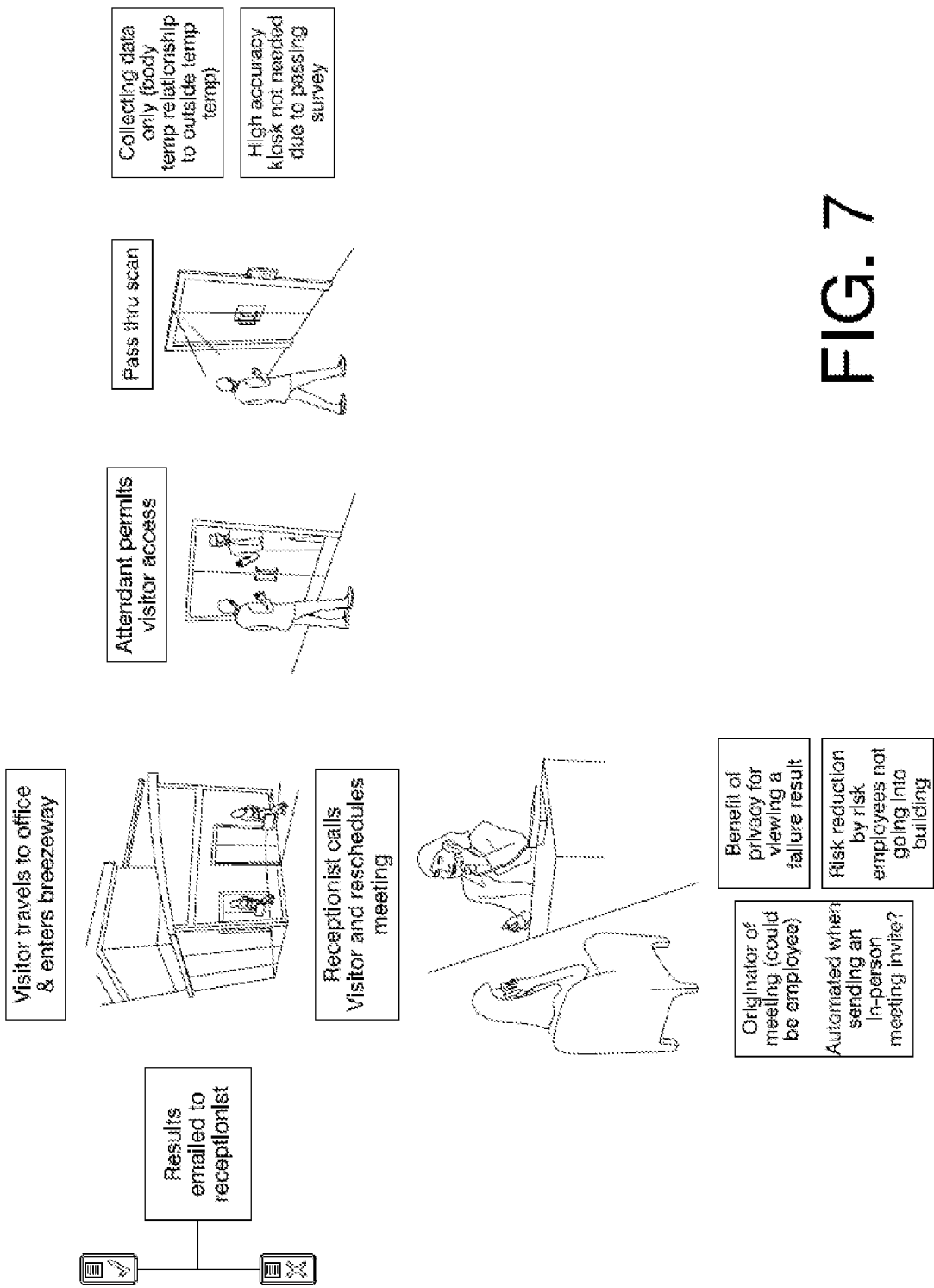


FIG. 7

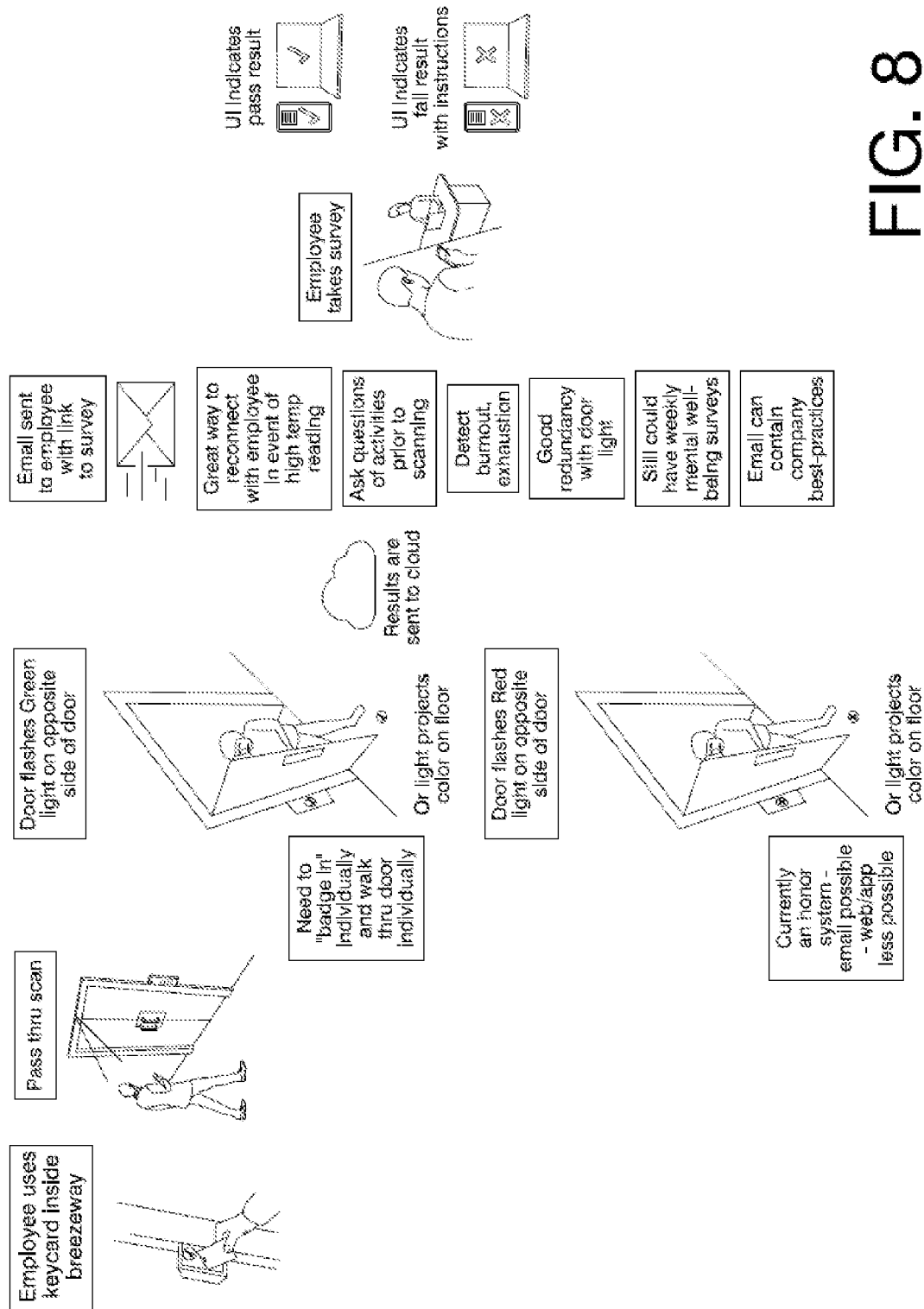


FIG. 8

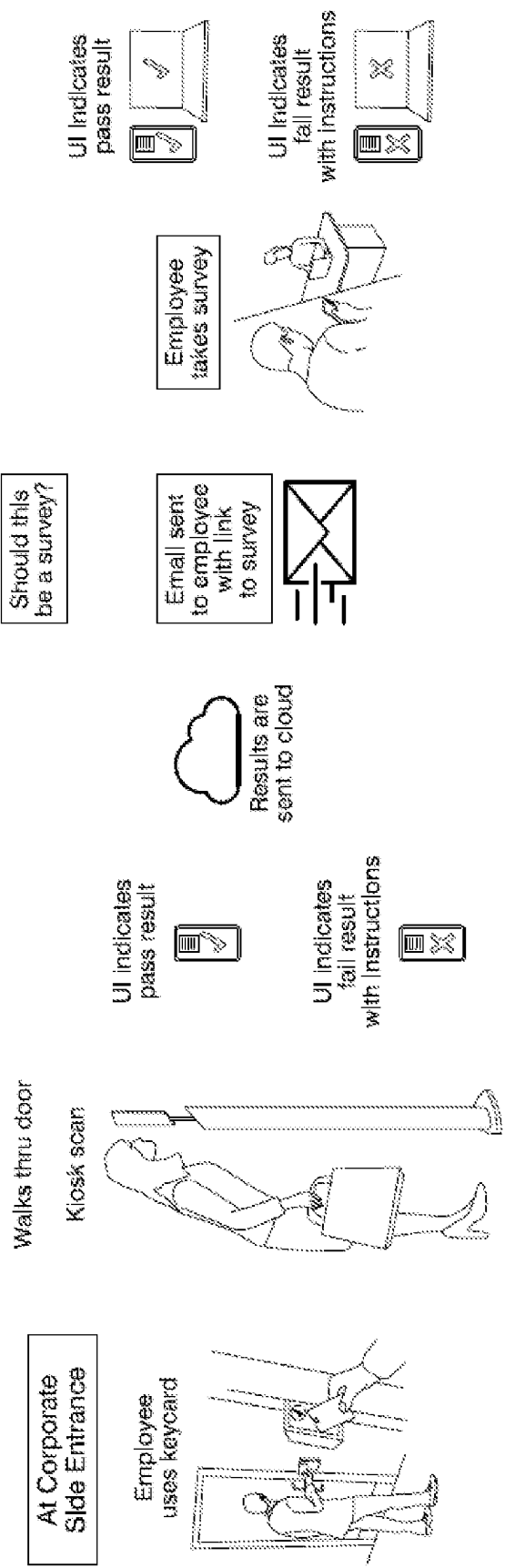


FIG. 9

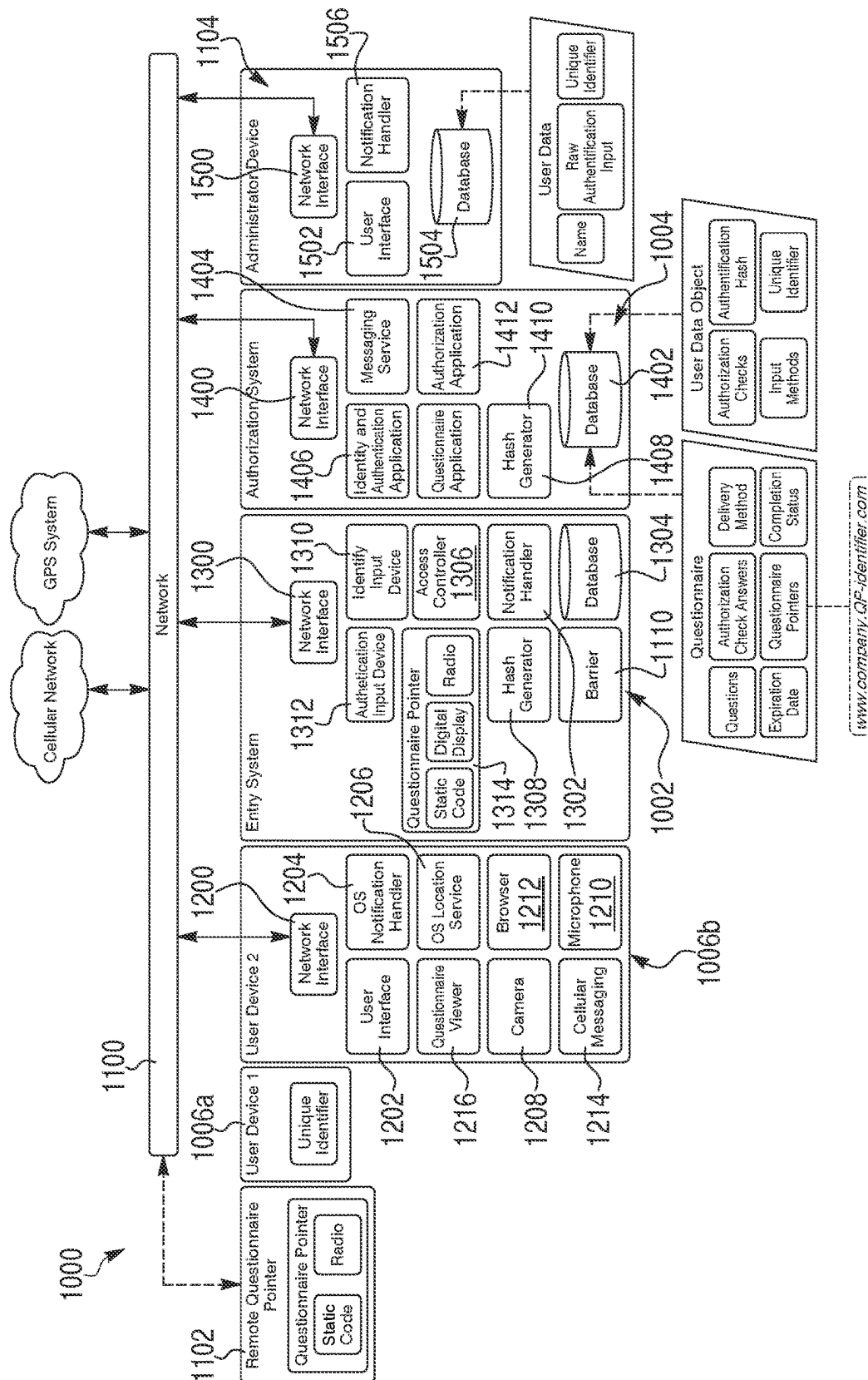


FIG. 10

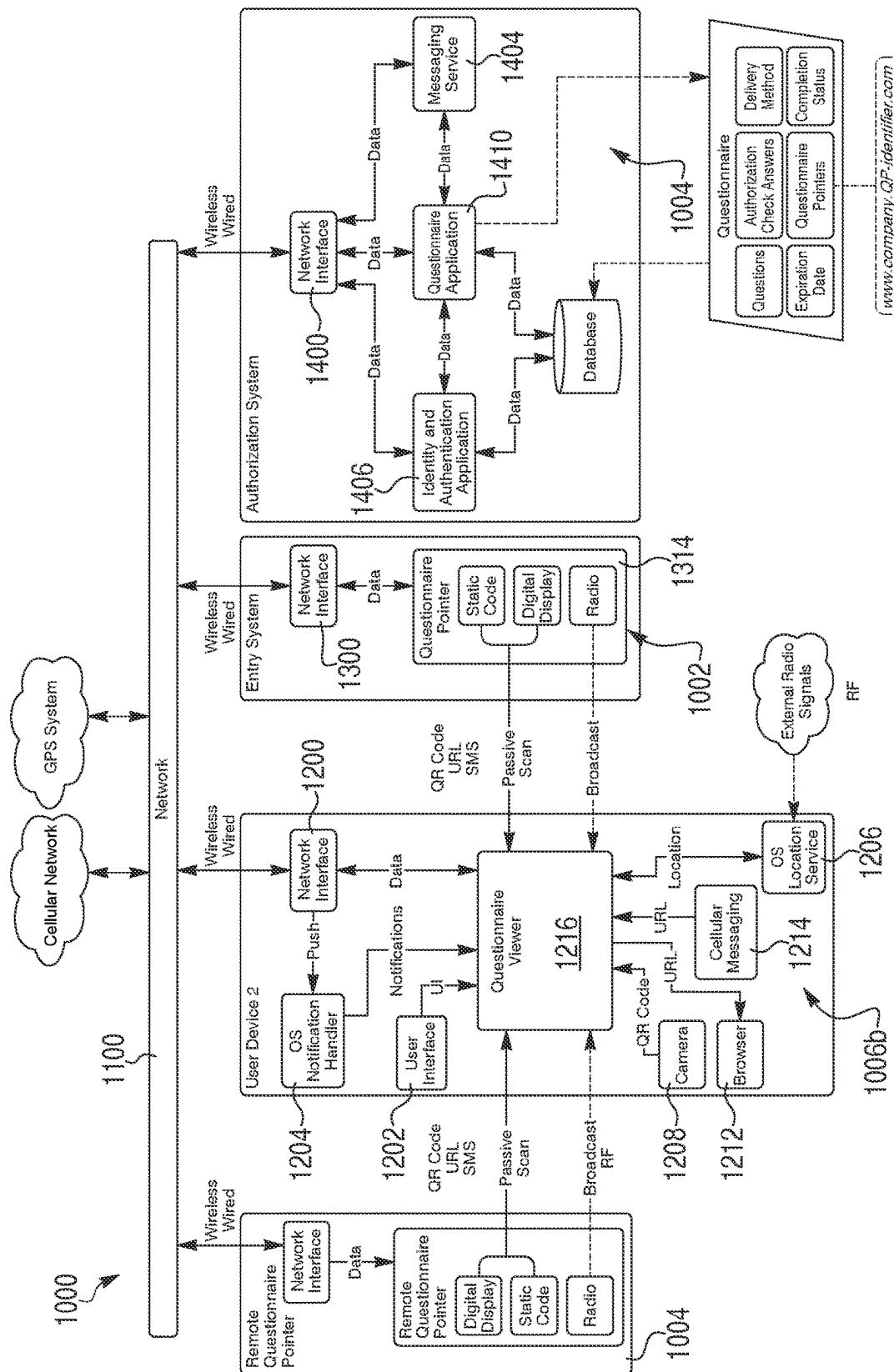
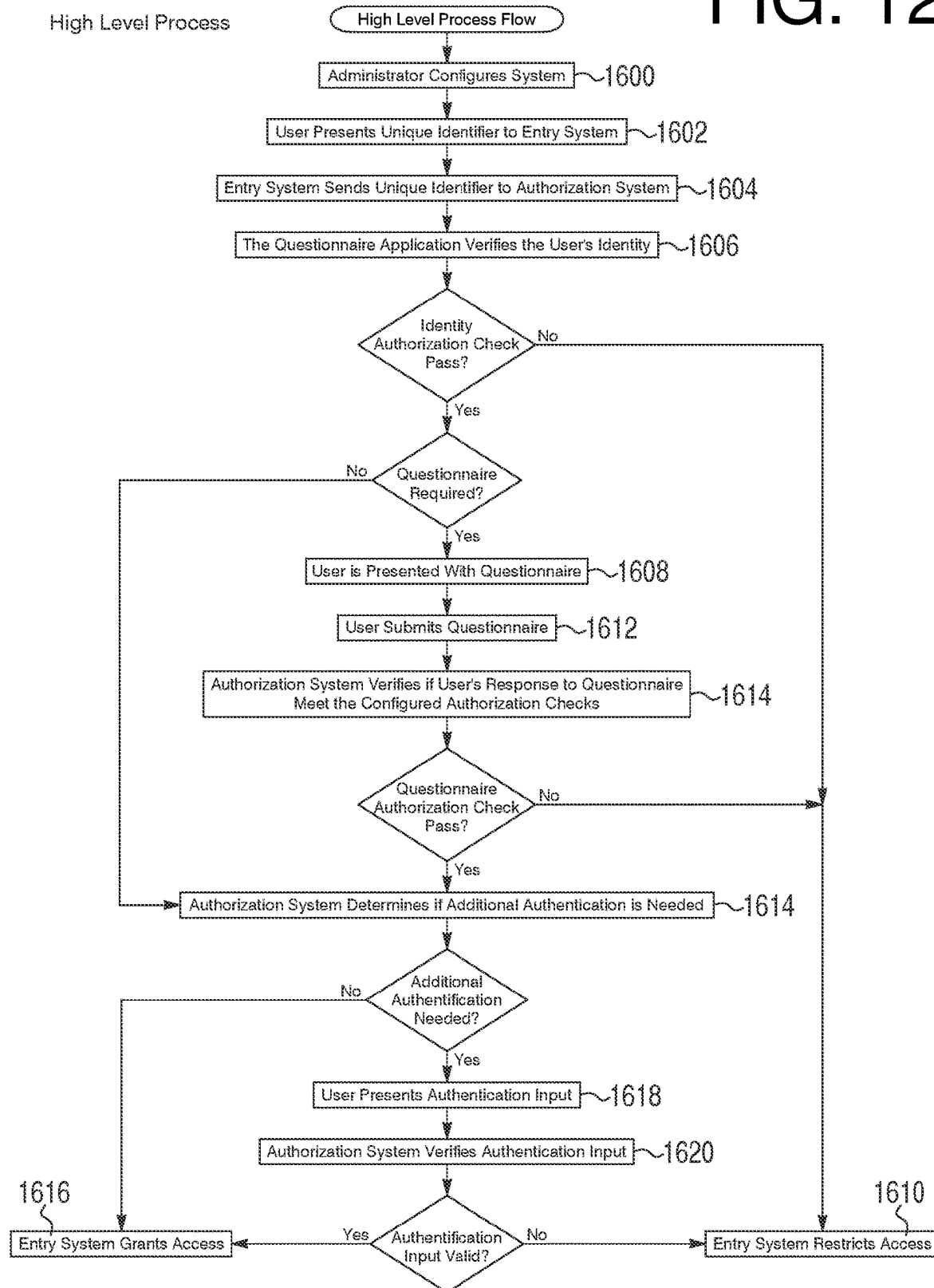


FIG. 11

FIG. 12



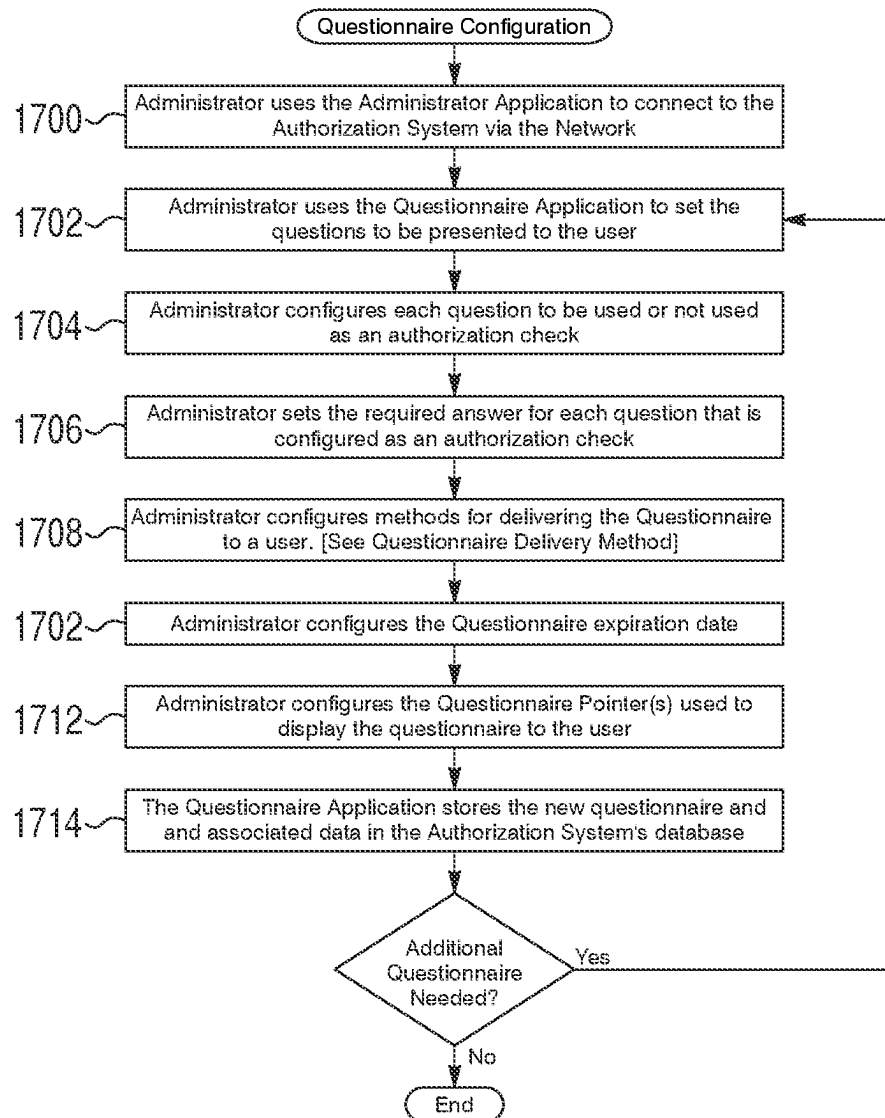


FIG. 13

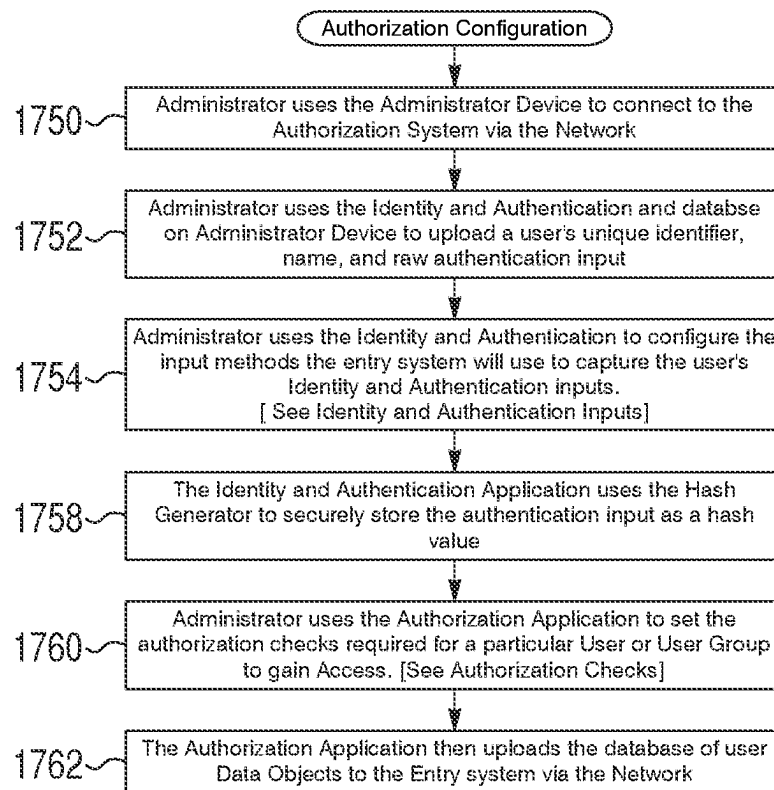
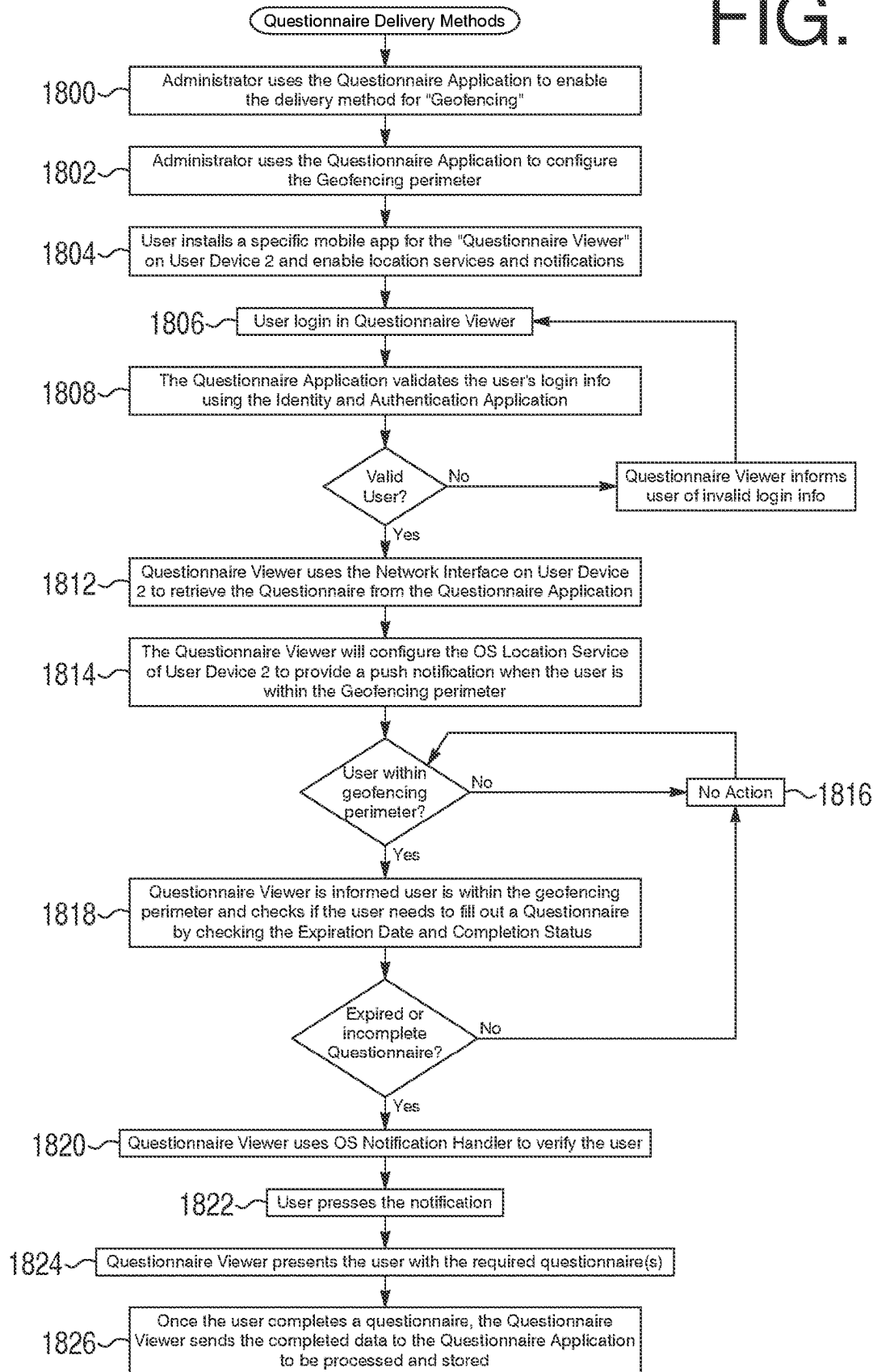


FIG. 14

FIG. 15



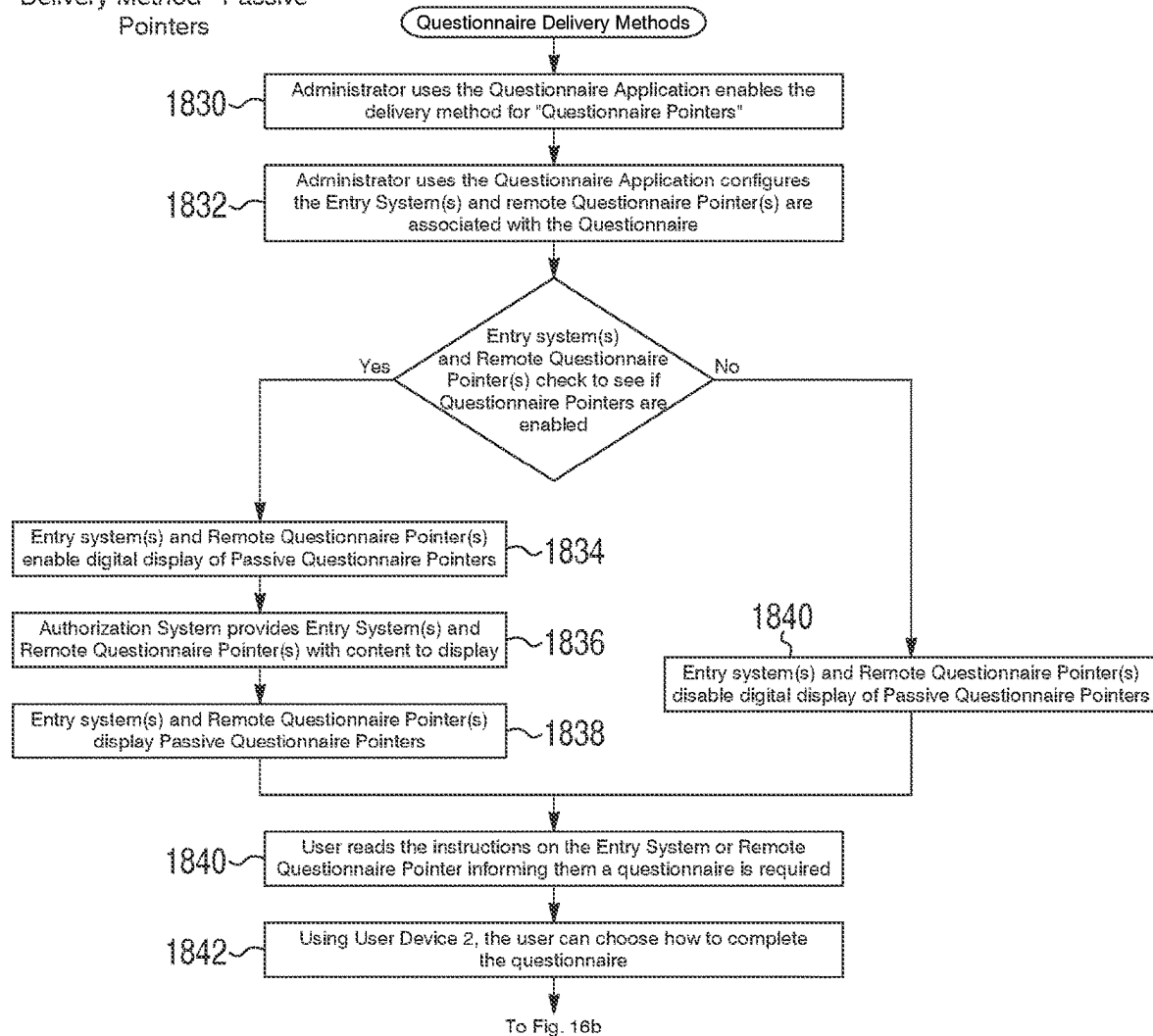
Delivery Method - Passive
Pointers

FIG. 16a

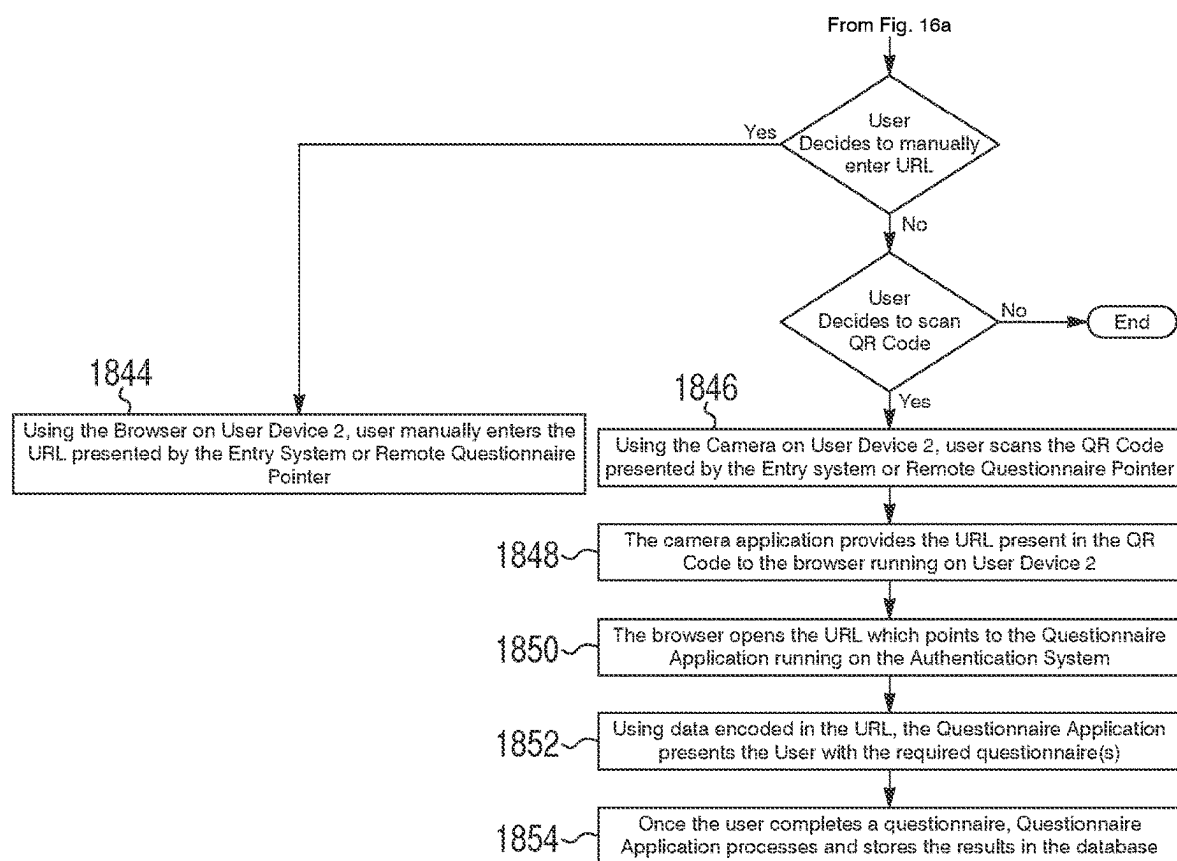
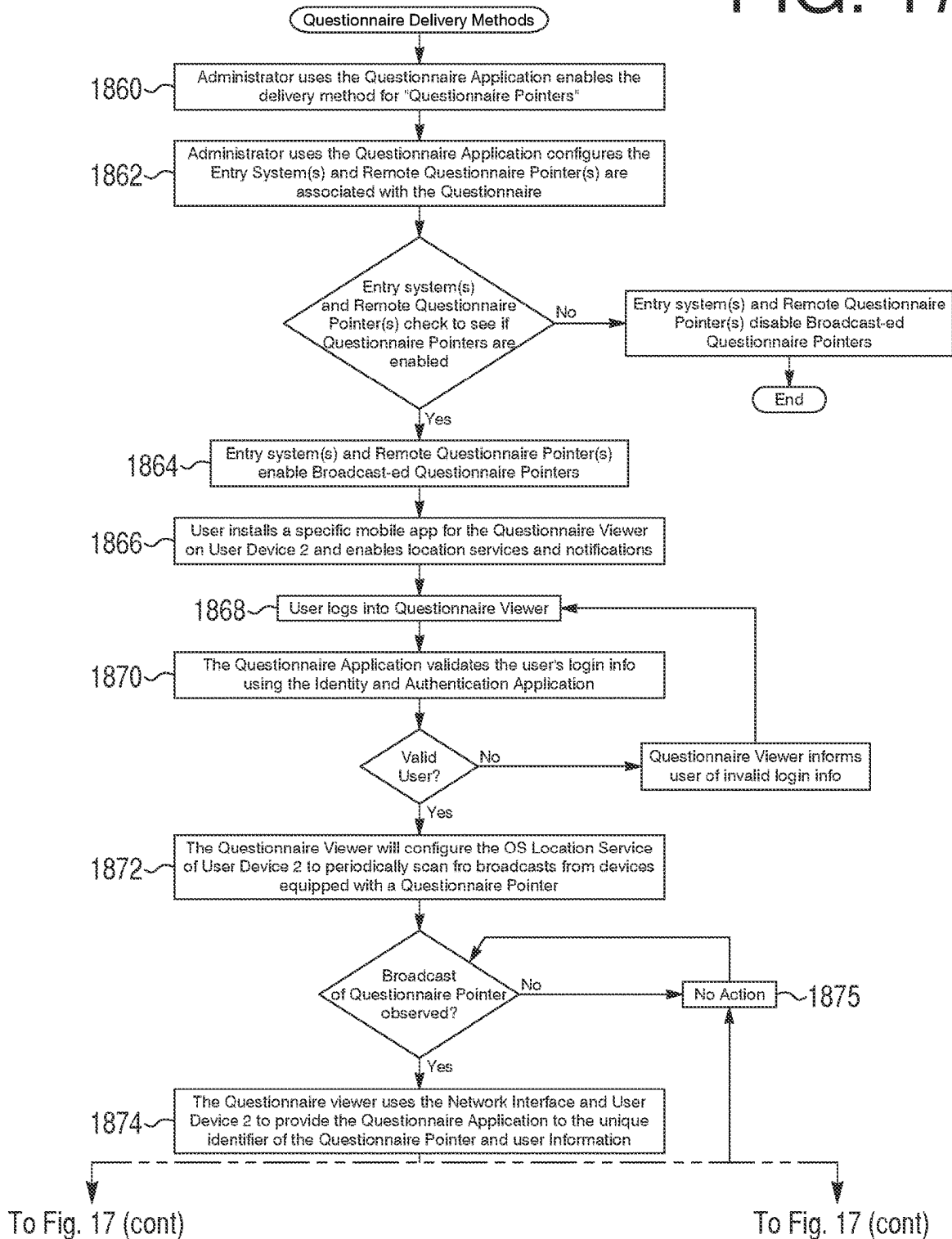


FIG. 16b

FIG. 17



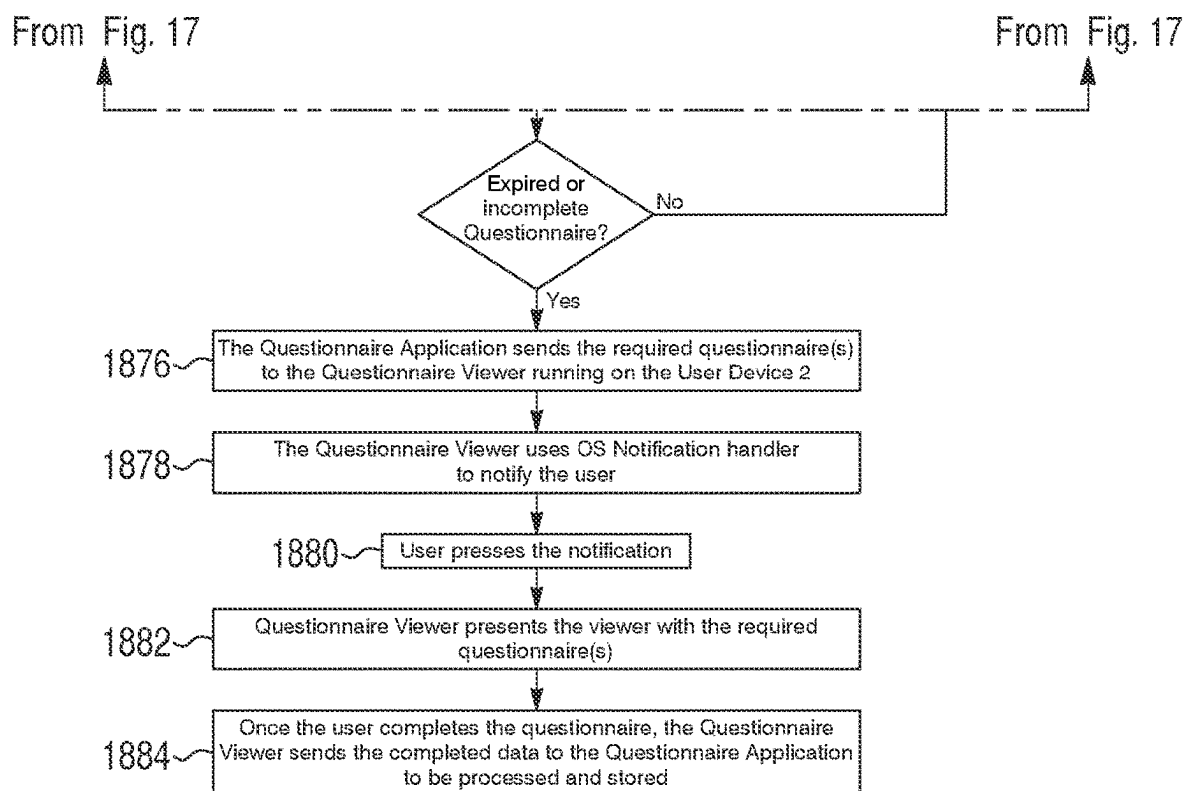


FIG. 17 (cont)

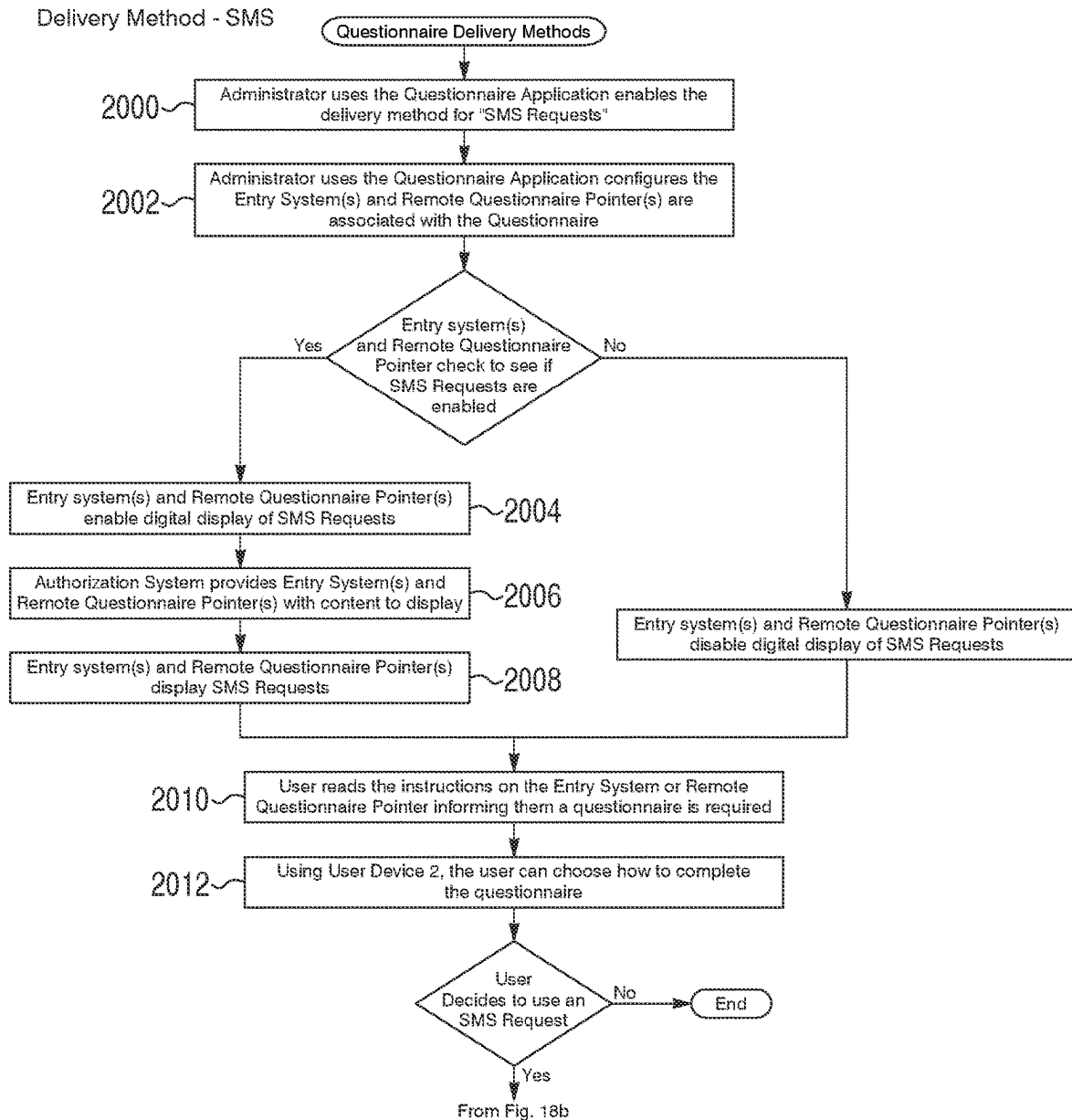


FIG. 18a

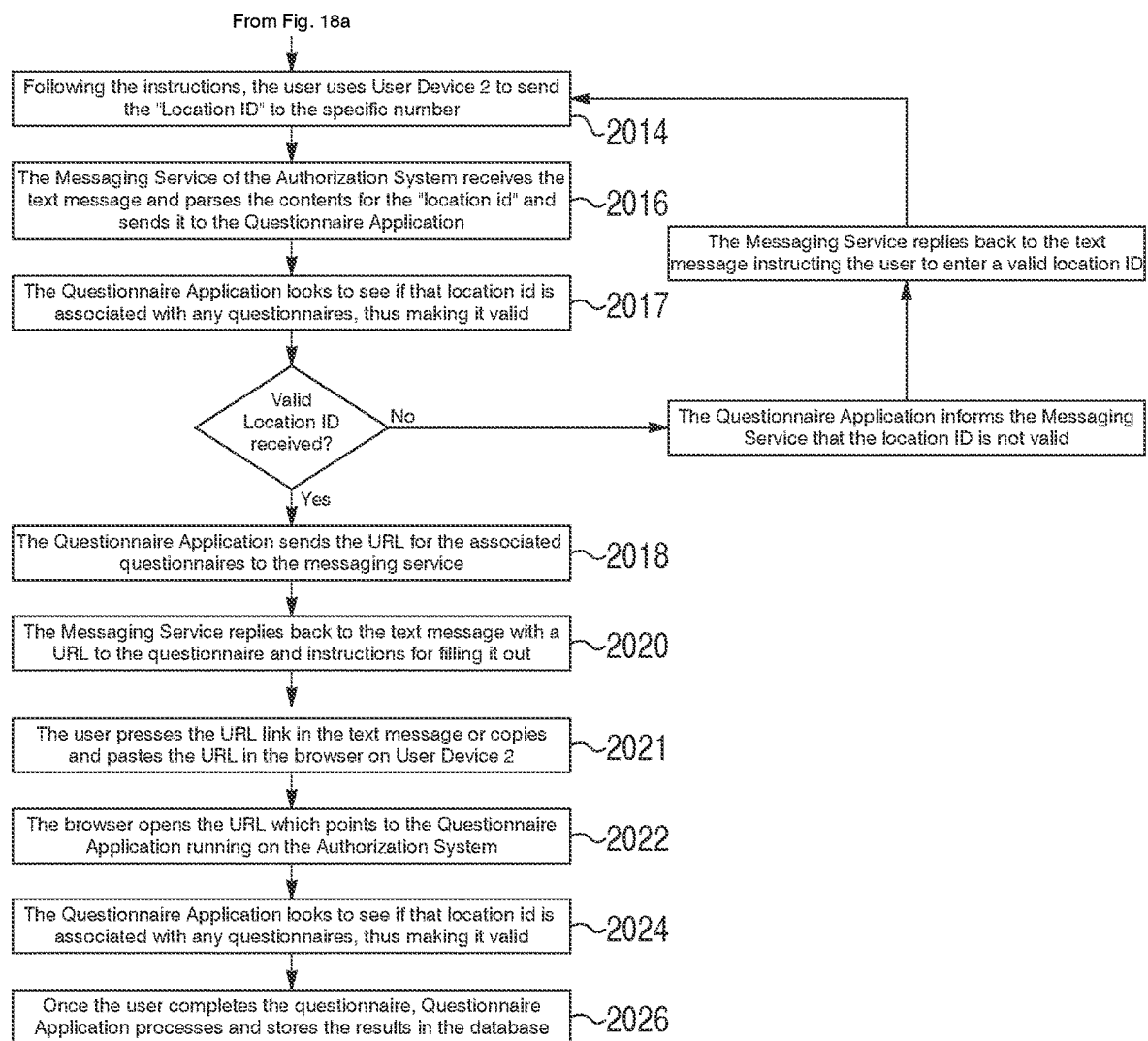


FIG. 18b

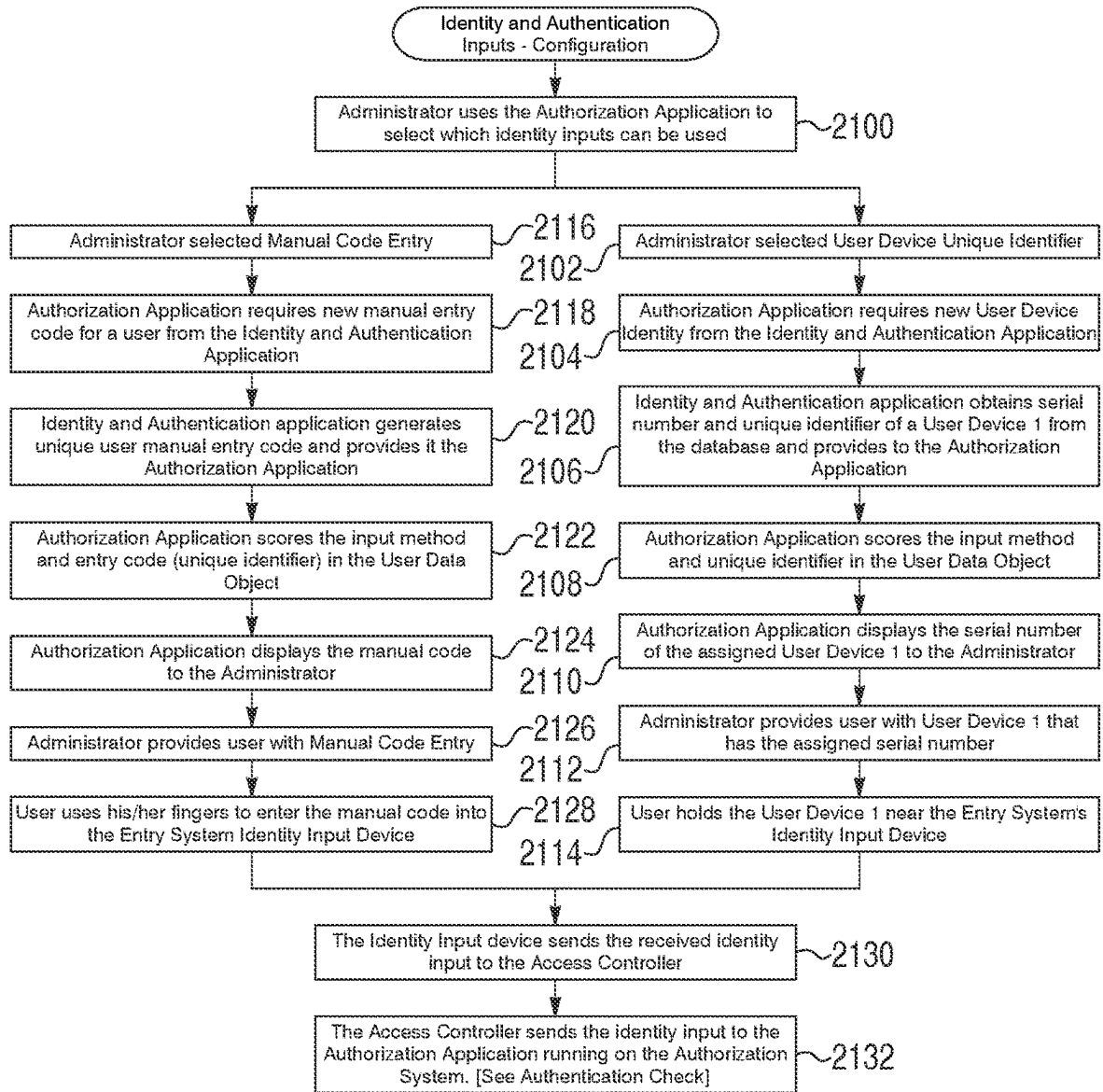


FIG. 19

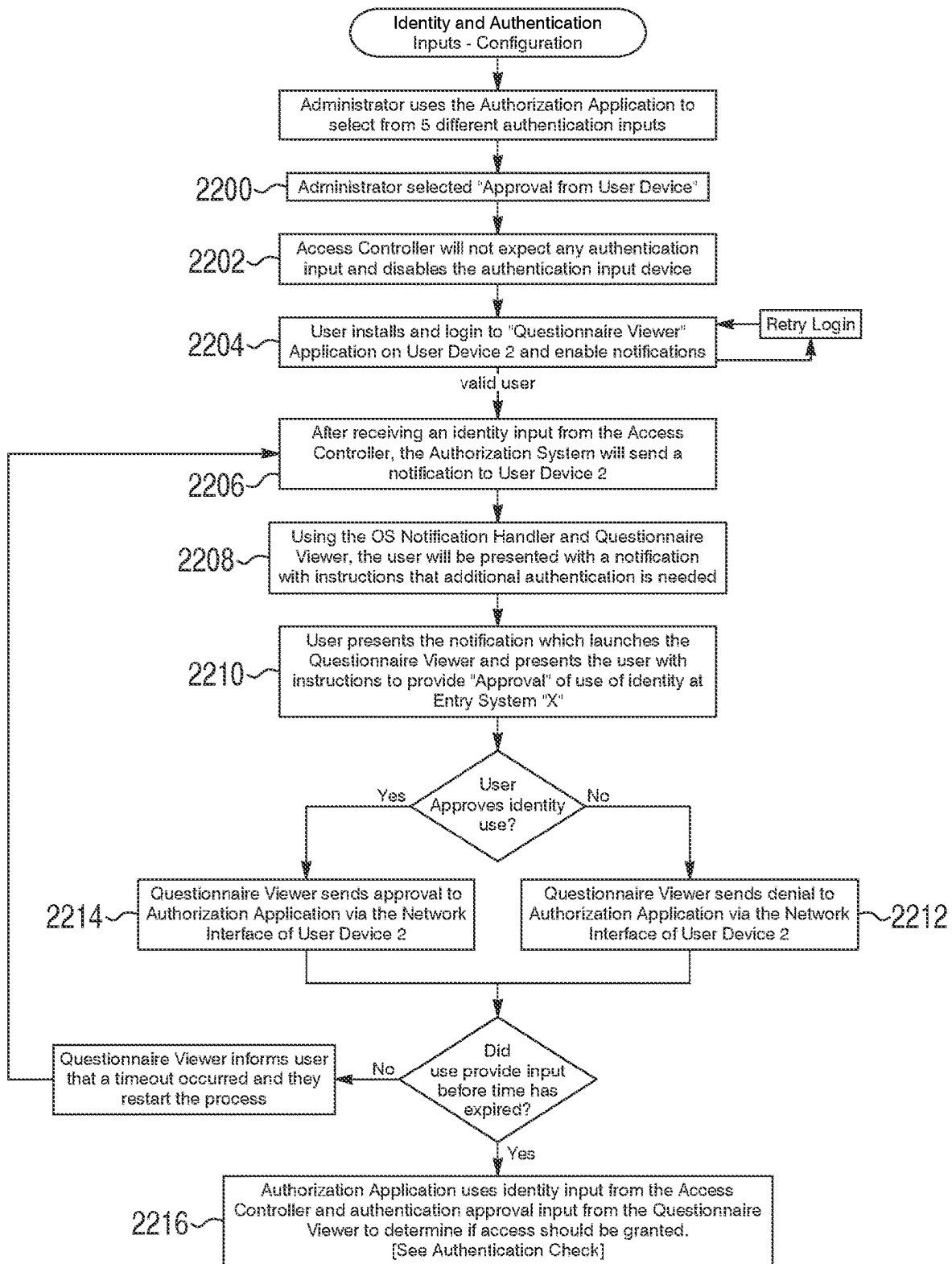
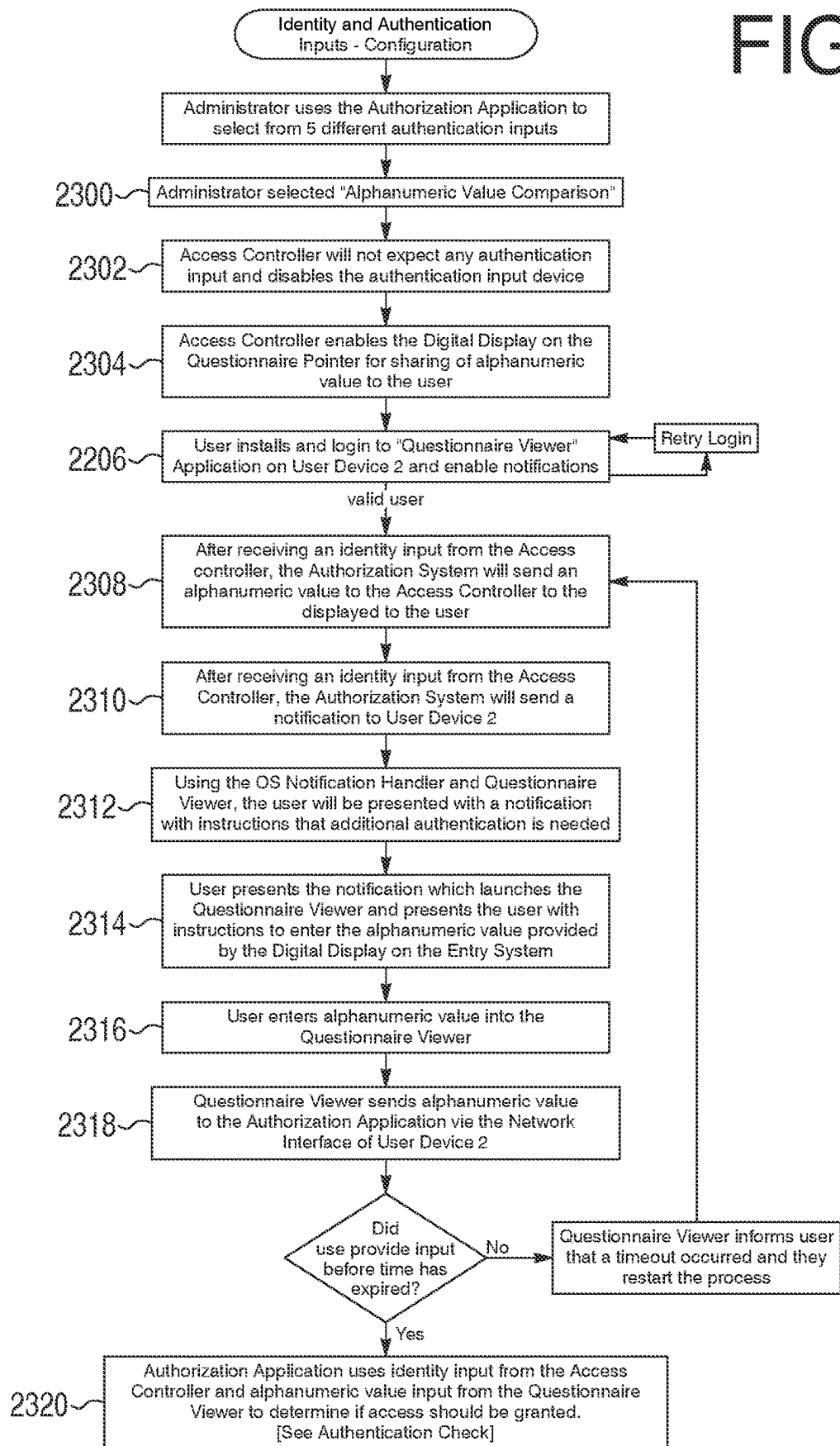


FIG. 20

FIG. 21



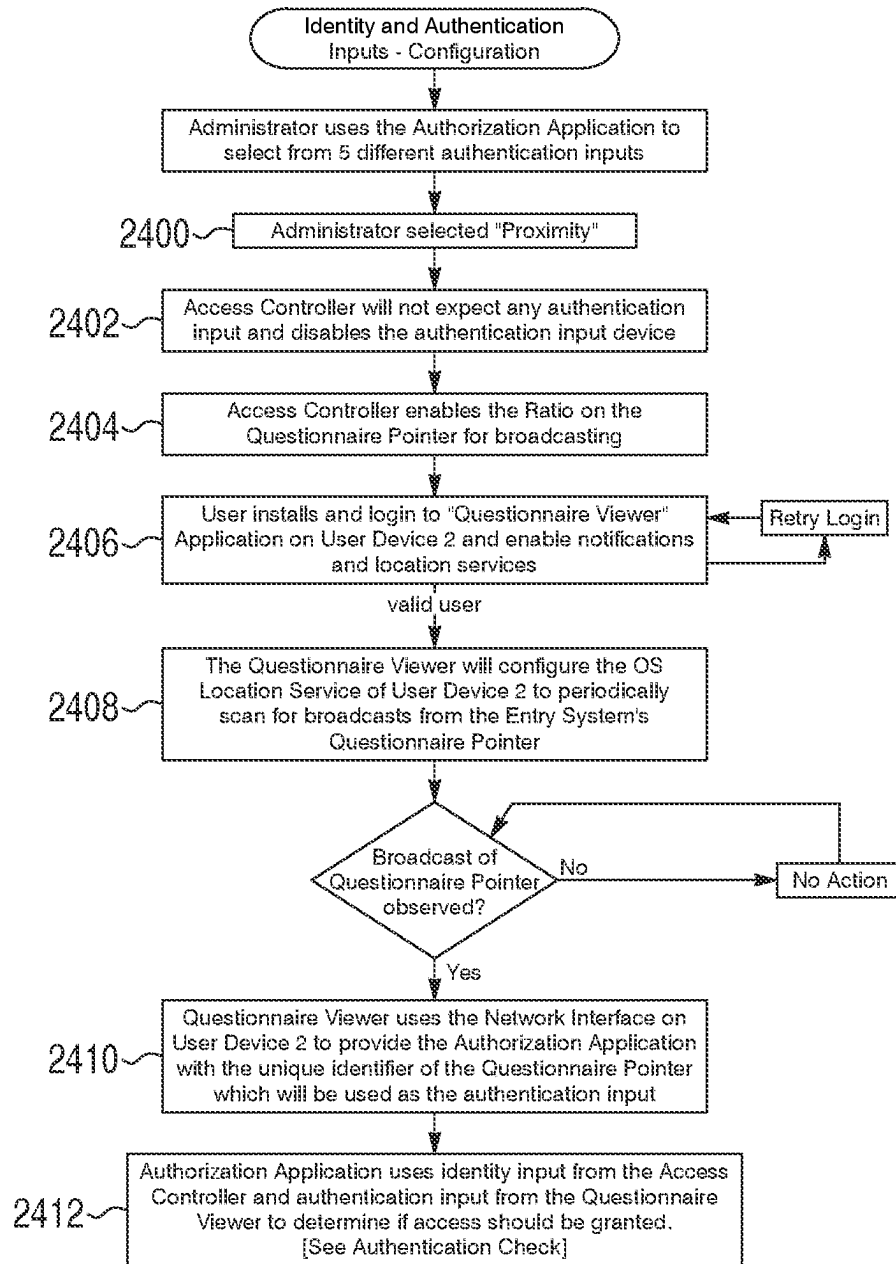


FIG. 22

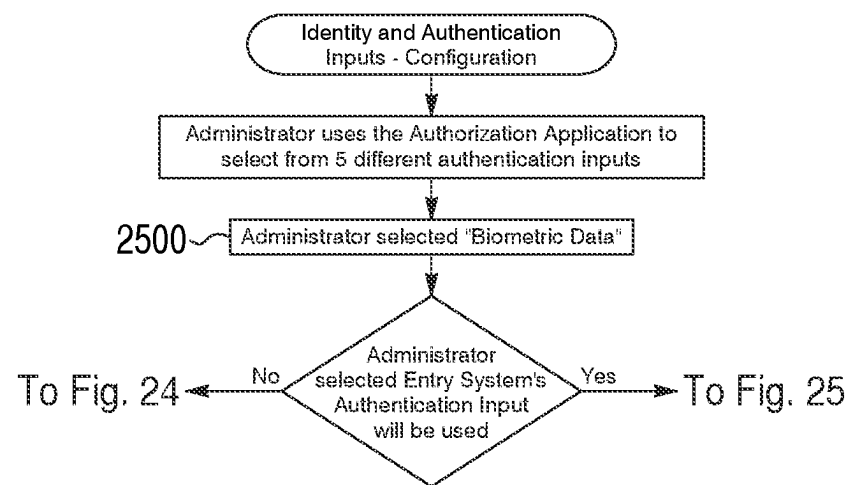


FIG. 23

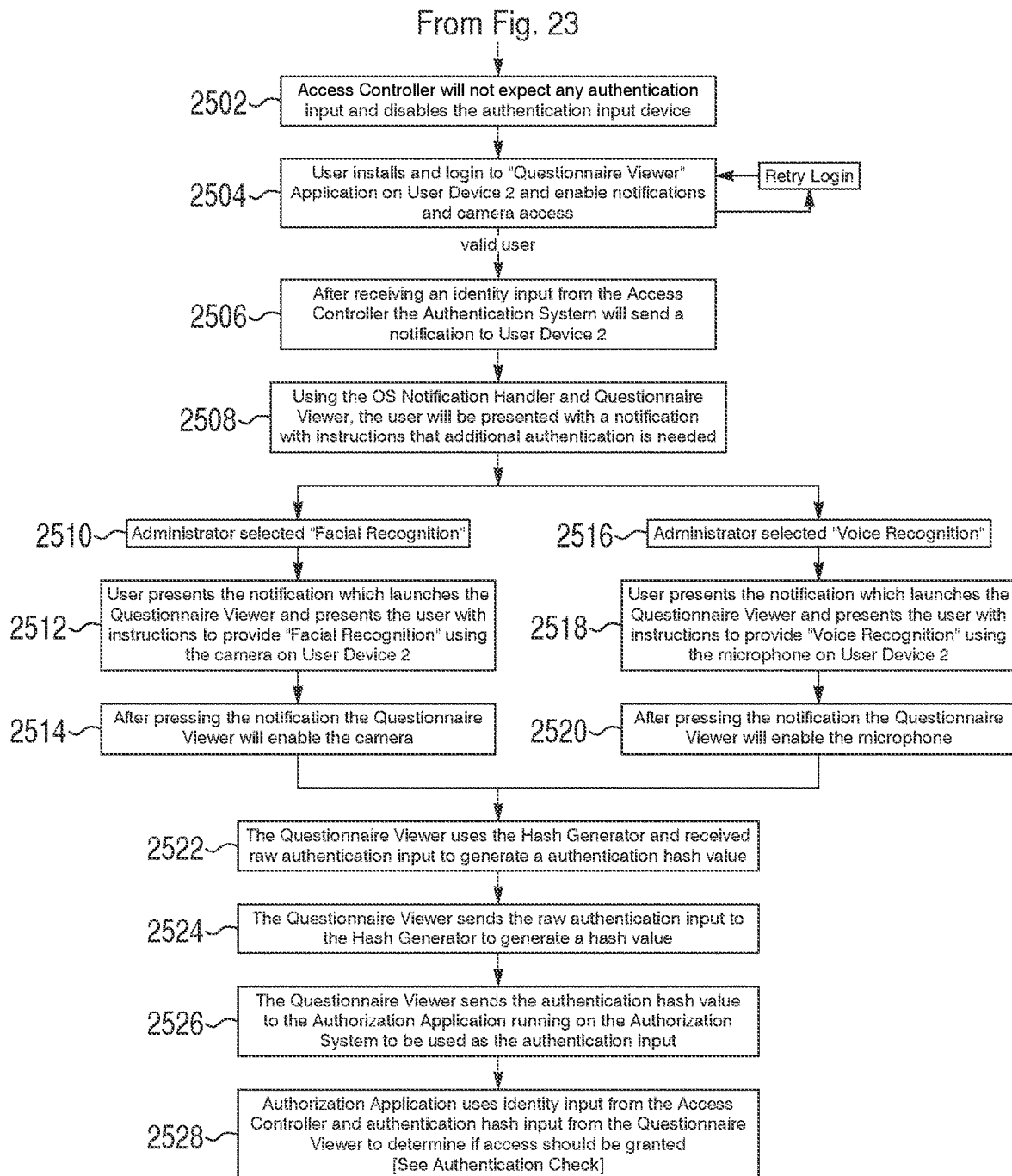


FIG. 24

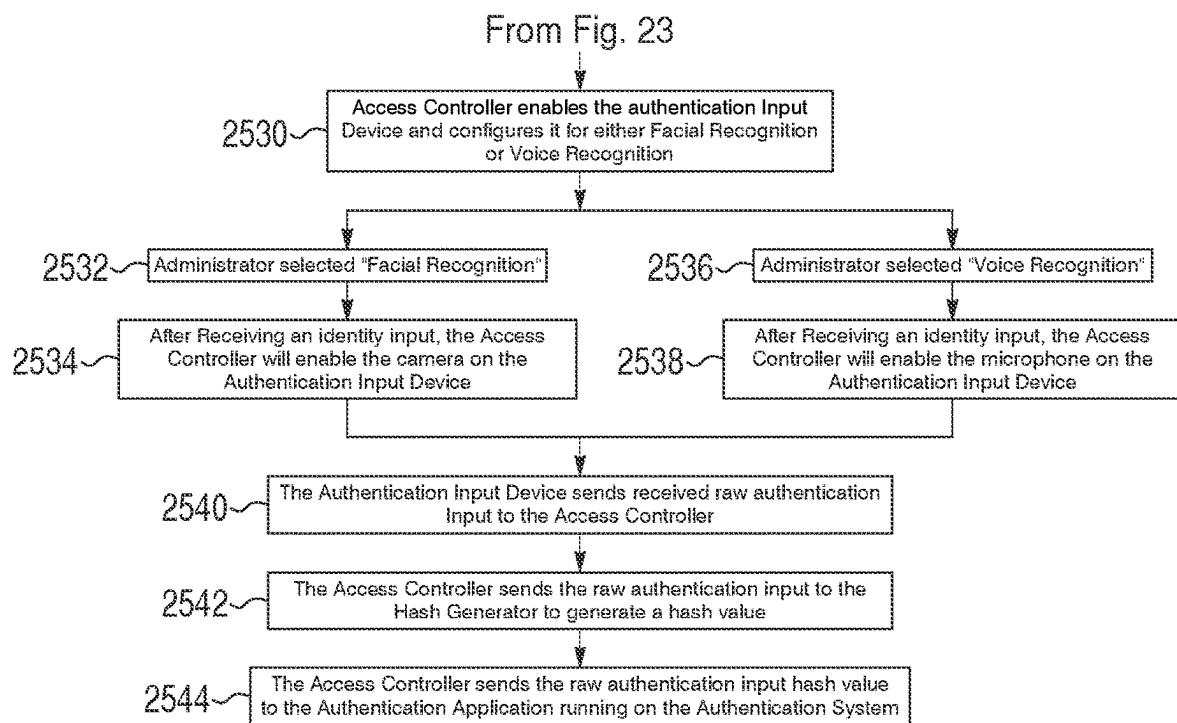
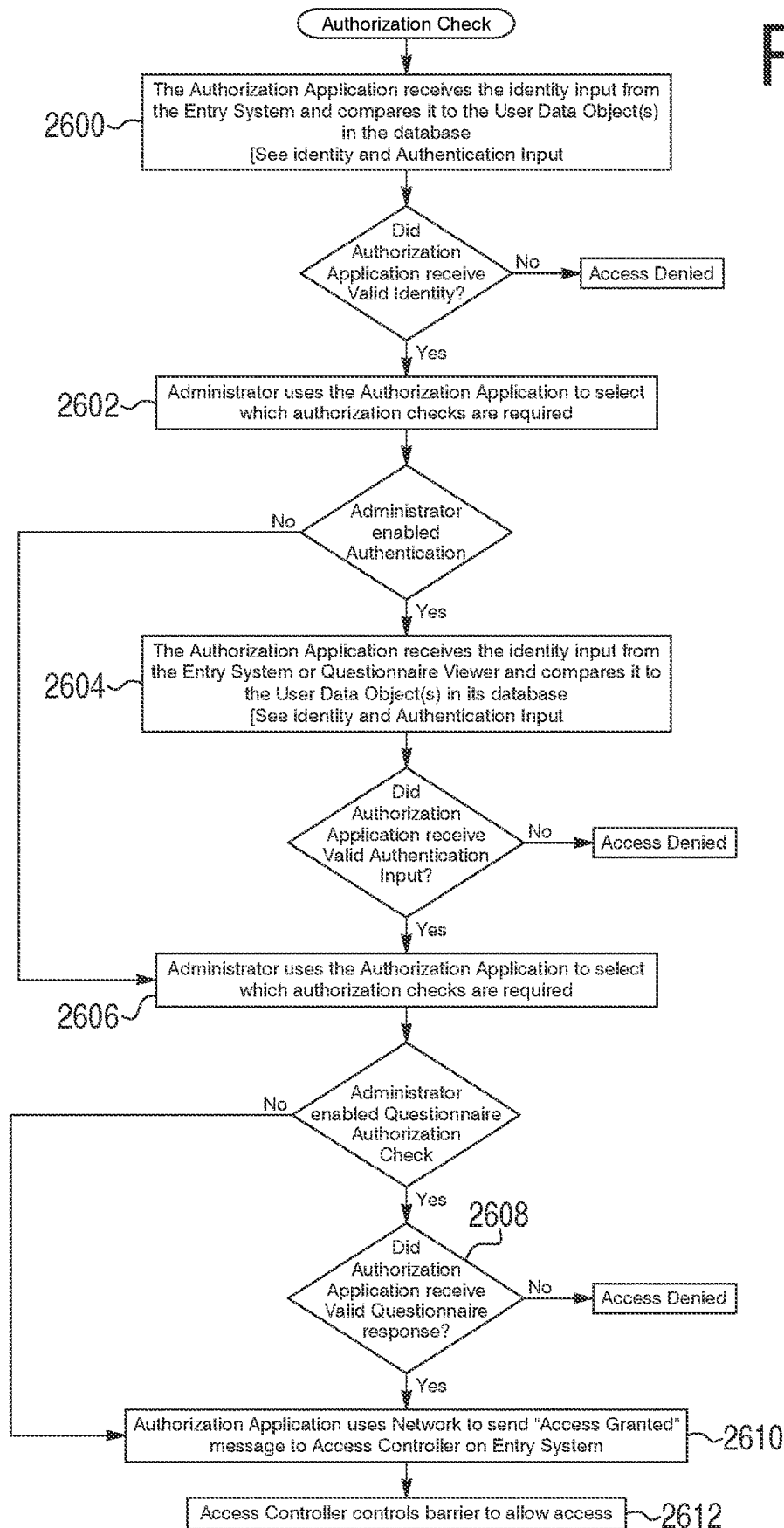


FIG. 25

FIG. 26



1

ENTRY ACCESS SYSTEM AND METHOD FOR QUESTIONNAIRE SCREENING

CROSS-REFERENCE TO RELATED APPLICATIONS AND CLAIM TO PRIORITY

This application is a continuation of U.S. patent application Ser. No. 17/473,492 filed Sep. 13, 2021, which claims the priority of U.S. Provisional Patent Application No. 63/077,193, filed Sep. 11, 2020, which are incorporated herein by reference.

FIELD OF THE INVENTION

The present invention relates to entry systems and methods containing identity verification and survey information, such as a wellness survey, security verification questionnaire, and the like, to control access to a buildings, secure areas or the like.

BACKGROUND OF THE INVENTION

With many contagious diseases isolation and social distancing are the most effective means to prevent the spread of the disease. For some diseases, such as COVID 19, a questionnaire may provide a rough indication of the risk of a person having the disease. For example, with COVID 19, contact with a disease positive individual, fever, respiratory symptoms, etc. may indicate that the person may have the disease. As such, office buildings sometimes question people for symptoms/risks of the disease to decide whether to allow entry. The entrants are usually required to answer the questionnaire at the door, either by a guard asking the questions or by filling out a form that is then reviewed before access is allowed. The process sometimes results in long lines at the door due to the time it takes to process the questionnaires.

Many buildings, such as office buildings or factories, already have entry systems which verify the identity of entrants before allowing entry, such as for security purposes. Such buildings would benefit from integrating a wellness questionnaire into the entry system, provided that long lines at the entrance were avoided. Therefore, there is a need for an entry system that provides identity verification as well as health screening monitoring and monitoring for regulatory, compliance and corporate governance of entrants prior to their entry into the building space.

Access to some buildings and facilities is sometimes limited for security and all regulatory requirements to individuals who have a certain credential that permit access. The credentials may be valid for a limited period, thus requiring periodic verification. Keeping track of individuals and whether they have the required credentials can be a laborious task, sometimes resulting a lines of individuals seeking access to a building or facility.

A system and method that keeps a record of individuals accessing a building or facility and also whether they have the necessary credentials in a time-saving manner is desirable in order to expedite entry and provide positive verification of who accessed the building or facility should the need arise.

SUMMARY OF THE INVENTION

The present invention provides an entry system for controlling access to a secure space, such as a building. In particular the entry system provides identity verification or

2

authentication, preferably multifactor identification, and a questionnaire/survey to control access to a building, secure area or the like. The questionnaire/survey may be used for health screening or to assure that individuals enter the secure space in a manner that assures regulatory compliance and/or compliance with internal corporate governance.

The entry system includes several components, such as a main entry system, a verification system, and a plurality of user devices that interact with the main entry and verification systems. The components preferably communicate with each other wirelessly to control access to the secure space.

Methods for making and using the different aspects of the present invention are also provided.

Other aspects of the invention, including apparatus, devices, kits, processes, and the like which constitute part of the invention, will become more apparent upon reading the following detailed description of the exemplary embodiments

BRIEF DESCRIPTION OF THE DRAWINGS

The following detailed description of the invention will be better understood when read in conjunction with the appended drawings. For the purpose of illustrating the invention, there is shown in the drawings embodiments which are presently preferred. It should be understood, however, that the invention is not limited to the precise arrangements and instrumentalities shown. In the drawings:

FIG. 1 is a schematic of the entry system of the present invention;

FIG. 2 is a perspective view of an exemplary entranceway;

FIG. 3 is a diagram illustrating the entranceway of FIG. 2 with geofencing technology;

FIG. 4 is a flow diagram showing the system as configured for use during an epidemic to control access by a user (employee) at a main entrance;

FIG. 5 is a continuation of the flow diagram of FIG. 4;

FIG. 6 is a flow diagram showing the system configured for use (to monitor general wellness or a specific disease or condition) to control access by a user (visitor) at a main entrance;

FIG. 7 is a continuation of the flow diagram of FIG. 6;

FIG. 8 is a diagram showing the system configured for use to monitor a user's (employee's) general wellness and to control access by the user at a main entrance;

FIG. 9 is a diagram showing the system configured for use to monitor a user's (employee's) general wellness and to control access by the user at a side entrance;

FIG. 10 is a diagram showing an exemplary entry system;

FIG. 11 is a diagram showing the data flow in the exemplary entry system for questionnaire delivery;

FIG. 12 is a flow chart showing operation of the exemplary entry system;

FIG. 13 is a flow chart showing the questionnaire configuration process;

FIG. 14 is a flow chart showing the authorization configuration process;

FIG. 15 is a flow chart showing a method for delivery of the questionnaire using geofencing;

FIGS. 16a and 16b are flow charts showing a method for delivery of the questionnaire using passive pointers (FIG. 16b is a continuation of FIG. 16a);

FIG. 17 is a flow chart showing a method for delivery of the questionnaire using broadcast pointers;

3

FIGS. **18a** and **18b** are a flow charts showing a method for delivery of the questionnaire using text messaging (FIG. **18b** is a continuation of FIG. **18a**);

FIG. **19** is a flow chart showing the identity input configuration process the user must follow to gain access to the secure space

FIG. **20** is a flow chart showing the authentication input using the second user device;

FIG. **21** is a flow chart showing the authentication input using alphanumeric value;

FIG. **22** is a flow chart showing the authentication input using proximity of the second user device to the main entry system;

FIG. **23** is a flow chart showing the authentication input using biometric data of the user;

FIG. **24** is a flow chart showing the authentication input using biometric authentication input from the user device;

FIG. **25** is a flow chart showing the authentication input using biometric authentication input from the authentication input device **1312** of the main entry system

FIG. **26** is a flow chart showing the authorization check process.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Reference will now be made in detail to exemplary embodiments and methods of the invention as illustrated in the accompanying drawings, in which like reference characters designate like or corresponding parts throughout the drawings. It should be noted, however, that the invention in its broader aspects is not limited to the specific details, representative devices and methods, and illustrative examples shown and described in connection with the exemplary embodiments and methods.

The present invention relates to entry systems for allowing entry into a secure space, such as an office building, a factory, a manufacturing facility, etc. Referring to FIG. **1**, the entry system **100**, preferably an automated system provided with a hardware and software platform, includes a main system **102** in electronic communication with an identity verification system **104** for verifying the identity of the building entrants, and interfacing with a plurality of user devices **106**, such as fob keys, smartphones, tablets, watches and the like.

Part of the identity verification system **104**, preferably includes a verification device **105**, as shown in FIG. **2**, that may be located at the entrance to the secure space, such as the entrance to a building, laboratory, etc. The entrance may include a frame **108** mounted to a wall around the entrance, and a barrier, such as a door, **110** pivotally or movably mounted on the frame **108**. The verification device **105** may be mounted to the frame or the barrier **110**. The barrier **110** can partially or completely cover the frame opening (FIG. **2** shows the barrier **110** partially covering the frame opening) and can be, but is not limited to, a door, a gate, a bar, a curtain, a turnstile, or combinations thereof. Other visual or physical restraints may also be appropriate for the present invention.

The identity verification device **105** may be, but is not limited to, an RFID reader, a fingerprint scanner, a retinal scanner, a camera, a keypad, a writing pad, a voice recorder, a bar code reader, a QR reader or combinations thereof. Identity (ID) verification may, e.g., include the capability of capturing images or data from or about individuals, and of comparing the image or data to data within existing internal

4

databases or third-party databases for identity matches. ID verification may be capable of capturing images of a user/person, or reading his/her identification card, employee badge, a bar code, a QR code, etc., and then cross referencing the images with the database of a third-party access control system (ACS). A photo of a person may be analyzed by a facial recognition algorithm to compare it against a database of images. Upon a match in the data or a positive identification of the individual person, the system may allow the person to enter the secure space.

In another embodiment, multi-factor verification may be used to improve the reliability of ID verification. Multi-factor verification or multi-factor authentication may be accomplished by requiring a user's entry privileges to be authenticated by using more than just a simple key card. At many secure facilities, users are provided with access badges to swipe against door readers to enter restricted areas. One problem with such access control is that anyone that has access to the badge can improperly gain access to the restricted area. Multi-factor verification minimizes this problem by validating the identity of the user that is requesting access twice, e.g., once with a key card or the like, and then a second time with facial recognition, fingerprints, iris scan, face/voice recognition, signature, or personal identification number.

A user device **106**, such as a smart phone, a smart watch, or a tablet is to be associated with the user. The user device **106** is provided with a hardware and software platform for electronic communication with the main system **102**. The user device **106** preferably communicates wirelessly with the main system **102**. The user device **106** communicates with the user interface electronically through electronic communication devices that may include sensors, gateways, switches, routers, hubs or any other electronic communication device to the remotely located main system **102**. The user device **106** may transmit its information, e.g., location and identity of the user, on an ongoing basis or intermittently (preferably intermittently), directly to the main system **102** or via the other electronic communication devices. Preferably, the user device **106** communicates with the main system **102** only when the user device is within specified areas (as discussed below). The communication preferably uses wireless technologies and protocols for electronic communication such as cellular (3G/4G/5G), Bluetooth, Bluetooth Low Energy, WiFi, TCP/IP, near field communication (NFC), and other such technologies and protocols.

In a preferred embodiment, geofencing may be used to trigger a push notification to the user device **106** advising the user to fill out a questionnaire before attempting to enter the secure space. The user device **106** communicates its location information to the main system **102** on some basis, such as periodic or intermittent. Referring to FIG. **3**, when the transmitted location is within a specified location or area **112** relative to the secure location **114**, the main system **102** is notified. The specified location **112** may be set in advance in the main system **102**, and may be, e.g., the parking lot of the secure space **114** and/or a specified distance from the secure space **114**. For example, as shown in FIG. **3**, the specified location may be an area within a radius R from the center of the secure space **114**. When a user is within the specified location **112**, as determined by the location of the user device **106**, the main system **102** sends a push notification to the user to fill out the questionnaire. The user may be instructed to log in to a website to answer the questionnaire or to answer the questionnaire sent directly to the user

5

device. Once the questionnaire is completed, it, along with the user identity information, is forwarded to the main system **102**.

In case of COVID-19, the questionnaire includes questions relating to risk factors indicating high risk of contracting the disease or symptoms of the disease. Of the symptoms, the questions may ask whether the user has one or more of the following symptoms:

- Fever or chills
- Cough
- Shortness of breath or difficulty breathing
- Fatigue
- Muscle or body aches
- Headache
- New loss of taste or smell
- Sore throat
- Congestion or runny nose
- Nausea or vomiting
- Diarrhea
- Loss of smell or taste

Other the risk factors include whether the user has

- Been tested for the disease
- Been diagnosed with disease
- Been in contact with a person with disease within the last 14 days
- Travelled to any regions affected by disease within the last 14 days
- Underlying health conditions, such as heart disease, lung disease, kidney disease, diabetes, autoimmune disorders, etc.

Other diseases, including other infectious diseases, mental health, or overall wellness, may also be similarly screened using similar questionnaires with risk factors and symptoms associated with that disease. The questionnaire may inquire about the user's mental state (angry/sad/happy), stress, or other conditions. For infectious diseases, the user should access the questionnaire before requesting entry to the secure space **114**. For general wellness and mental health, the user should access the questionnaire periodically, once a week or once a quarter.

While we disclose use of the system **100** for health screening, it may be used to assure that individuals enter in a manner that assures regulatory compliance and compliance with internal corporate governance. The system **100** utilizes observed data (biometrics) and user-provided data to determine whether to perform some action, such as allow access, track wellbeing, provide a notification, etc. Security regulations may, for example, preclude entrance where an individual has visited a country or region within a period of time. Corporate governance may preclude entrance until an entrant has signed a non-disclosure agreement (NDA). The system **100** allows multiple types of questionnaires to be provided to a proposed entrant in order to assure compliance with health, regulatory and corporate governance protocols. Regulatory and corporate governance may include updating or providing appropriate non-disclosure agreements (NDAs), security clearance, safety protocols, codes of conduct, non-compete, engagement census, and other re-accruing processes that require acknowledgment or feedback from employees or visitors.

In addition to the questionnaire, the system **100** may include one or more sensors **116** (FIG. 2), such as a temperature sensor, a camera, other the like. For example, the sensor **116** may be a temperature sensor to detect body temperature of the user. Alternatively, the sensor **116** may be a camera with associated software to detect the user's facial features to determine the user mood (angry/sad/happy).

6

Once transmitted to the main system **102**, the questionnaire is valid only for a selected amount of time. For example, the selected amount of time may be 10 minutes or less, 5 minutes or less, preferably about 2 minutes. The selected amount of time is preferably calculated to allow the user to arrive at the entrance of the secure area to request entry. While health screening may require that the period during which a questionnaire response is valid is relatively short, other protocols, such as for regulatory and corporate governance may allow a much longer period, such as a calendar quarter or possibly longer. After the selected amount of time has passed, the answered questionnaire expires. If the user is still in or near the specified location **112**, the main system **102** notifies the user device **106** of the expiration and requests that another questionnaire be completed. Alternatively, the main system **102** may notify the user device **106** that the questionnaire is expiring and request that the user affirmatively extend the expiration.

Operation of the entry system **100** is described in the following. When the user is within the specified location **112**, the user device **106** notifies the main system **102**, e.g., via an application installed on the user device **106** that is in communication with the main system **102**. The user is then advised to fill out the questionnaire. Alternatively, the user may fill out the questionnaire before leaving for the secure space **114**. One advantage of filling out the questionnaire before leaving is that the user is informed whether he/she is cleared for entry before having to drive all the way to the secure space **114**. Once completed, the questionnaire and user information are transmitted to the main system **102**. Preferably, if the questionnaire is not cleared, the user is immediately notified that entry will not be permitted. When the user first requests entry into the secure space **114** and presents his/her credential, such as a badge containing an RFID chip, the identify verification device **104** reads the badge or other credential and confirms the identity of the user. At the same time, the main system **102** confirms whether the user has filled out the questionnaire. The person's authorization to enter is validated only when the person's identity and questionnaire are cleared. Once validated, the person is granted access and entry to the secure space **114**. To be granted access to the secure space **114**, the person's identity must be verified, and the questionnaire must be cleared, i.e., no risk or symptom of COVID-19 (or other infectious disease). If one or more of the symptoms or one or more risk factor are present, entry may be denied. If entry is denied, in certain embodiments, a receptionist/attendant/healthcare professional is called to perform further testing, perform further questioning, and/or to reschedule entry for a later time. The receptionist/attendant/healthcare professional may meet the user at the entrance to the secure space **114** or contact the user at a later time.

When epidemic disease condition is not prevalent, the system **100** may be used to monitor the user general well-being. For example, the system **100** may be used to detect the individual's general wellness. If unwell conditions are detected by the system **100**, reminders may be sent to the user to take a day off and/or to seek healthcare.

FIGS. 4-5 illustrate use of the system **100** when an employee wishes to access the main entrance of the office or like work facility. In that event, the employee/user may take the survey from home and access to the facility will be permitted when the survey is successful and a key card or like identifier swiped at the entrance determines the identity of the employee and that the employee has successfully passed the survey. If the survey has failed, then access is denied.

FIGS. 6-7 illustrate use of the system 100 when a visitor seeks to access the facility. In that event, the visitor's identity is first determined, such as by presentation of a valid driver's license or other government identification card, and then the survey completed successfully prior to access being allowed.

FIGS. 8-9 illustrate use of the system 100 when being used to monitor the general wellness of employees. The operation is similar to FIGS. 4-5, except that the survey questions relate to the general well-being of the individual and are not directed to a particular disease.

FIG. 10 shows an exemplary entry system 1000 of the present invention utilizing multifactor verification. The system 1000 includes a network 1100 operating over cellular, WiFi, and/or wired Ethernet networks and GPS system. The system 1000 includes first user devices 1006a, second user devices 1006b, a main entry system 1002, a verification (or authorization) system 1004, and an administrator device 1104. FIG. 11 shows the communication and data flow of the system 1000.

The network 1100 preferably uses wireless technologies and protocols for electronic communication such as cellular (3G/4G/5G), Bluetooth, Bluetooth Low Energy, WiFi, TCP/IP, near field communication (NFC), and other such technologies and protocols. Preferably, the network 1100 also provided with location information from the second user device 1006b via the GPS on the second user device 1006b. The components of the system communicate with each other via the network 1100.

The system 1000 preferably uses two user devices 1006a and 1006b for two-factor authentication. The first user device 1006a may preferably be, but not limited to, a FOB key, a card key, or other devices using RFID technology, for example, to enter a building. The first user device 1006a need not interface with the network 1100, as it may be read by the verification system 1004 at the entrance to the secure space 114 or on the pathway to the secured space 114. Preferably, the first user device 1006a provides a unique identifier for every user, so that the main entry system 1002, on reading the first user device 1006a, identifies the first user device 1006a as associating with a particular user.

The second user device 1006b may preferably be a smart phone, a smart watch, laptop, or a tablet associated with the user. The second user device 1006b preferably includes one or more of the following modules: a network interface 1200, a user interface 1202, an operating system (OS) notification handler 1204, a location service 1206, a camera 1208, a microphone 1210, a browser 1212, a cellular messenger 1214, and a questionnaire viewer application 1216. A typical smart phone typically includes all of those modules.

The network interface 1200 interfaces the second user device 1006b to the network 1100 wirelessly, such as by cellular, WiFi, and/or wired Ethernet networks, and is usually provided by the operating system of the second user device 1006b. The network interface 1200 is used by the device 1006b to communicate with the main entry system 1002 and the verification system 1004 via the network 1100.

The user interface 1202 is preferably provided by the OS of the second user device 1006b and provides a platform for the user to input information to applications running on the second user device 1006b.

The notification handler 1204 is provided by the OS of the second user device 1006b and handles the receiving and presentation of push notifications to the device 1006b (see FIG. 11). The notification handler 1204 is used by the questionnaire viewer 1216 to let the user know he/she has a questionnaire to fill out (see FIG. 11).

The Location Service 1206 provides location information to applications running on the second user device 1006b. It is preferably provided by the second user device 1006b and may use GPS, WiFi, Bluetooth, and other triangulation methods for providing the location of the second user device 1006b. The questionnaire viewer 1216 preferably uses this location service 1206 in two ways: 1) to automatically present the user with a questionnaire based on the location or proximity to an entry system; and 2) to use the location service 1206 as an authentication input as it provides additional assurance that a particular user is the one who is attempting to gain access (multifactor authentication).

The camera 1208 is preferably provided by the second user device 1006b and operable via the device's OS. The user may use the camera 1208 to scan a QR Code to launch a questionnaire. Additionally, the questionnaire viewer 1216 can also use the camera 1208 to provide an authentication input in the form of an image of the user's face, e.g., for facial recognition for use in multifactor authentication (see FIG. 11).

The microphone 1210 is preferably provided by the second user device 1006b and operable on the device's OS. The questionnaire viewer 1216 can also use this to provide an authentication input in the form of a recording of the user's voice, e.g., for voice recognition for use in multifactor authentication.

The browser 1212 is preferably provided by the second user device 1006b and operable on the device's OS. This browser can be used by the user to fill out the questionnaire.

The cellular messenger 1214 may be a text message (a.k.a. SMS) service provided by the second user device 1006b and operable on the device's OS. A user may use this cellular messenger 1214 to request a URL to a questionnaire from the system 1000.

The questionnaire viewer 1216 is preferably an application on the second user device 1006b that the user uses for filling out and managing questionnaires. It can also configure the OS to provide notifications and location tracking used by the questionnaire viewer 1216 to automatically present users with the questionnaire.

The main entry system 1002 allows entry into the secure space 114. It preferably includes processors and associated electronics that are purpose-built for the application, such as the access controller, questionnaire pointer, notification handler as explained below; however, off-the-shelf electronic components, such as the barrier, authentication input device, identity input device, network interface, and database as explained below, may be included. The main entry system 1002 preferably includes one or more of the following modules: a network interface 1300, a notification handler 1302, a database 1304, a barrier 1110, an access controller 1306, a hash generator 1308, an identity input device 1310, and an authentication input device 1312.

The network interface 1300 interfaces the main entry system 1002 to the network 1100 wirelessly, such as by cellular, WiFi, and/or wired Ethernet networks. The network interface 1300 allows the main entry system 1002 to communicate with the network 1100, and thereby, with the second user device 1200 and the verification system 1004.

The notification handler 1302 allows the main entry system 1002 to provide user feedback (i.e., notifications). These notifications may provide reminders to the user of actions need by the user. For example, the notifications may let the user know that a questionnaire will expire in the near future or has expired but is not needed for access. Additionally, for health screening, the notifications may indicate to the user whether he/she has a health metric that is out of the

normal range. Other notification actions may be providing an NDA, additional security clearance, health screening, safety protocols, codes of conduct, non-compete, engagement census, or other re-accruing processes that require acknowledgment or feedback from employees or visitors.

The database **1304** is a storage device used by the access controller **1306** to store system configuration data, history logs, user data objects, and other data required for proper operation.

The barrier **1110** provides a physical barrier used to grant and restrict access to the secure space **114**. A typical barrier **1110** is a door having a latch for locking and unlocking of the door. The access controller **1306** electronically controls the locking and unlocking of the latch to allow or restrict access to the secure space **114** via the barrier **1110**.

The access controller **1306** is preferably an electronic device equipped with a processor, memory, storage, and software application(s). The access controller **1306** receives local inputs from the authentication input device **1312** and the identity input device **1310**. In addition, the access controller **1306** communicates with other devices on the network via the network interface **1300**. The access controller **1306** is responsible for controlling the barrier **1110** which controls access to a physical space. Furthermore, the access controller **1306** may be used to control the QP **1004** and notification handler **1302**.

The hash generator **1308** (a.k.a. hashing function, cryptographic hashing algorithm) is used to convert and securely store sensitive biometric (e.g., facial imagery and voice) data. The hash generator converts data of an arbitrary size to data of a fixed size. This compresses the data to significantly reduce the amount of storage required to store the data. It also reduces the amount of data that has to be shared over the network **1100**. A cryptographic hash algorithm for the hash generator provides additional security and privacy benefits since the user's raw biometric data is never stored. The hash generator **1308** may be a separate module (as shown in FIG. **10**) or on the same electronic board as the access controller **1306**.

The identity input device **1310** is used to capture the identity input from the user. A preferred implementation involves an RFID scanner for reading the first user device **1006a** or a keypad for entry of an entry code (e.g., a combination of letters and/or numbers uniquely assigned to the user) from the user. This input is then sent to the access controller **1306**.

The authentication input device **1312** captures an authentication input from a user. The authentication input device **1312** may include a digital camera and/or a microphone that can be enabled/disabled by the system administrator via the configuration of the system, e.g., for multi-factor authentication. The authentication input device **1312** captures raw imagery of a user's face and/or voice, which are sent to the access controller **1306** and eventually to the authorization system **1004** to authenticate a user through facial or voice recognition as explained below.

The questionnaire pointer (QP) **1314** is a sign or display, preferably controlled by the access controller **1306** and provides a link to a questionnaire required for accessing the building. The QP **1314** preferably provides a URL link to a questionnaire/survey/form that the user must fill out before entry into the secure space **114**. The QP is preferably an electronic sign that provides the URL link by providing a QR Code or phone number for receiving the URL text message. Additionally, the QP **1314** can be used for authentication purpose. The QP **1314** preferably includes three modules: a digital display, a static code (e.g., a non-digital

sign), and a radio. The digital display allows the main entry system **1002** to display text and imagery that are electronically sent from the access controller **1306**. When used for delivering a questionnaire to a user, the display can be configured to display a QR Code, URL, or phone number (see FIG. **11**). These allow the user to access the questionnaire/survey/form that the user is required to complete, preferably via the second user device **1006b**. When used for authentication purposes, the digital display can provide a one-time-use alphanumeric value to the user. The user would then use the questionnaire viewer **1216** on the second user device **1006b** to enter this value. Providing a correct alphanumeric value authenticates that a particular user is attempting to use a specific entry system at a particular point in time.

The static code is used to provide a URL link to a questionnaire/survey that visitors of a building must fill out before entry. The QP **1314** can provide the URL link by giving a static QR Code or phone number for receiving the URL via text sent to the second user device **1006b**. The radio is electronically controlled by the access controller **1306**. When used for delivering a questionnaire to a user, the radio can broadcast a URL that can be scanned by a user's phone. This can be done via Bluetooth, Bluetooth Beacons, or NFC. Additionally, the radio may be a WiFi radio, which is configured as an access point with no password required. However, when the user attempts to join the network, the user is presented with a questionnaire instead of a login screen. Once the user submits the questionnaire, the user's device will be removed from the network (the user cannot use this network to connect to the internet). When used for authentication purposes (multifactor authentication), the radio can be used to determine whether the second user device **1006b** is within close proximity, preferably less than about 100 m, more preferably less than about 50 m, most preferably less than about 20 m to the barrier **1110** where the identity code is entered by the user. In this scenario, the radio broadcasts itself to be scanned by the second user device **1006b** via the questionnaire viewer **1216**. The questionnaire viewer **1216** then sends this information (whether the second user device **1006b** is proximate to the barrier **1110**) to the authorization system **1004**. If the second user device is indeed proximate to the barrier **1110**, the authorization system **1004** authenticates the user who entered the identity input information into the main entry system **1002**.

The verification system (authorization system) **1004** preferably includes one or more servers that can be installed on-premise or remotely. For latency concerns, local installation is preferred. However, a remote server may provide a less expensive alternative. The authorization system **1004** is used for hosting, managing, and configuring a questionnaire application **1410** located thereon. Additionally, the system determines whether a particular user is authorized to enter a building or area within a building. The authorization system **1004** verifies the identity and authentication inputs that the user presented to the main entry system **1002** and/or the second user device **1006b**. Furthermore, the authorization system **1004** may require the user to submit a questionnaire before authorizing access by the user. The authorization system **1004** takes the user's inputs and determines whether or not the user has presented the correct information to be authorized for access. The authorization system **1004** preferably includes one or more of the following modules: a network interface **1400**, a database **1402**, a messaging service **1404**, an identity and authentication application (IAA) **1406**, a hash generator **1408**, the questionnaire application **1410**, and an authorization application **1412**.

11

The network interface **1400** interfaces the authorization system **1004** to the network **1100** wirelessly, such as by cellular, WiFi, and/or wired Ethernet networks. This network interface **1300** allows the authorization system **1004** to communicate with the network **1100**, and thereby, with the second user device **1006b** and the access controller **1306** on the main entry system **1002**.

The database **1402** is a storage device used by the authorization system **1004** to store system configuration data, history logs, user data objects, questionnaires, questionnaire submissions, and other data required for proper operation.

The messaging service **1404** allows the user to send a text message to request a link to the required questionnaire. The user may be notified to access the questionnaire, e.g., by a notification push to the second user device **1006b** or by instruction from the authorization system **1004** or the main entry system **1002**. The user is instructed, e.g., by the QP, to provide his/her location by indicating the position ID in the body of the text. The messaging service **1404** parses the user's text for the location identification (ID). With the location ID the messaging service will then send this to the questionnaire application (see below) to determine which QP is associated with that location ID and obtain a URL to those questionnaires. The messaging service **1404** then sends the questionnaire(s) link(s) back to the user, preferably via SMS, with link(s) to the questionnaire(s) associated with that location (see FIG. 11).

The identity and authentication application (IAA) **1406** is configured to manage new identity and authentication inputs, and to ensure that new identity inputs (manual codes or unique identifiers of the first user device) are unique and never re-used. Additionally, The IAA **1406** also handles the secure storage of biometric data used for authentication purposes (see FIG. 11).

The hash generator **1408** (a.k.a. hashing function, cryptographic hashing algorithm) is configured to convert and securely store sensitive biometric data. The hash generator takes data of arbitrary size to data of a fixed size. This compresses the data to significantly reduce the amount of storage required to store data. It also reduces the amount of data that has to be shared over a network. A cryptographic hash algorithm for the hash generator provides additional security and privacy benefits, as the user's raw biometric data is never stored on the system. Although FIG. 10 shows the hash generator **1408** as a separate module, it may also be part of the IAA **1406**.

The questionnaire application **1410** is configured for the creation, management, and hosting of questionnaires. Administrators can remotely access the application **1410** for managing the questionnaires and export submissions for data-keeping purposes. The user can access the questionnaires hosted by the questionnaire application **1410** via a URL or directly in the questionnaire viewer running on his/her second user device **1006b**. The questionnaire as described above and provides a list of questions used to aid in health screening, regulatory, compliance, and/or corporate governance. The questionnaire is preferably saved on the database **1402**. The questions from the questionnaire are requested to be filled out by the user. Each or some of the questions may be configured to be used as an additional authorization check for determining authorization. The user's response for question(s) used for authorization check must match a list of acceptable answers (provided by the administrator at time of questionnaire creation) to receive authorization to enter the secure space **114**. Otherwise, authorization is denied. The questionnaire may be delivered

12

to the user in different ways. The system **1000** may disable/enable functionalities based on the deliver method selection based on the system configuration for the particular user. The delivery of the questionnaire may be through geofencing as discussed above. When the second user device **1006b** is within a pre-defined perimeter, preferably within about 50 m the secure space **114**, any required questionnaire(s) is automatically sent to the second user device **1006b**. The location is determined by the OS location services running on the second user device **1006b**. The location may also be determined by WiFi, Bluetooth, GPS, or other triangulation methods. The delivery of the questionnaire may also be through the QP **1314** or the remote questionnaire pointer (RQP) **1102** (as describe below). The QP **1314** or RQP **1102** are devices that provide a questionnaire URL to the user. Depending on the capabilities of the device that is providing the URL, there are two types: passive QP and broadcast QP. Passive QPs require the user to manually perform an action to get a link to the URL. For example, the user can scan a QR code, request a URL via SMS, or type in a URL displayed by the device. Broadcast QPs use devices equipped with radios which broadcast the URL. The questionnaire viewer **1216** running on the second user device **1006b** receives the radio broadcast when it is within radio range, and notifies the user that a questionnaire must be completed to access the secure space **114**. The radio broadcast may be implemented with WiFi, Bluetooth, or NFC. The user's responses expire on after a predetermined amount of time or when the questions are no longer in use, whichever comes first. The predetermined amount of time depends on the particular question and may be from about 1 day to about several years. For example, for a COVID 19 questionnaire, the expiration may be daily. On the other hand, for an NDA, the expiration may be determined by the language of the NDA. A list of devices used as QPs (RQPs **1102** and QPs **1314**) associated with a particular questionnaire are also saved in the database **1402**. The URL provided to the user preferably contains an identifier to allow the authorization system **1004** to know which device is used to obtain the URL. If the device is not associated with a questionnaire, as set by the system configuration, the user is not presented with a questionnaire. A list of users who have completed the questionnaire is also saved on the database **1402**.

The authorization application **1412** is used to configure the authorization checks required for a particular user or user group. The configuration may be done by the system administrators. The authorization application **1412** creates and maintains the user data object saved on the database **1402**. The authorization application **1412** uses the user data object to grant access to the user. When the user requests access, the main entry system **1002** only grants access if the authorization application **1412** determines all of the user's authorization checks have been met. The user data object is unique to each particular user and used by the authorization system **1004** to determine authorization access, and inform the system how the user can provide authentication/identity inputs to the system. The particular information contained in the user data object is explained below. The authorization system **1004** also maintains a list of the authorization checks required for each particular user to gain access to the secure space **114**. For all users, however, identification is always required to gain access. Two other authorization checks may also be required of the user to gain access: an authentication input (to authenticate the user entering the identity input) and/or questionnaire responses. The additional authentication may be facial or voice recognition, for example. If additional authentication is required, the hashed value of the

raw authentication input (facial or vocal), if that input method is selected, is generated and saved in the database **1402**. The authorization application **1412** may also be used to define input methods for each particular user. Depending on the particular user, the authentication and identity input methods may differ. For example, a CEO may be required to have a more secure method of authenticating themselves when compared to a janitor. Depending on the user, no authentication may be necessary or one or more input methods may be required (depending on the level of security required to enter the secure space **114**).

One authentication input method is approval from the second user device. Here, when the user uses his first user device **1006a**, e.g., at the barrier **1110** or his assigned code (as described below), the user is presented with a notification that their identity is being used at the particular harrier **1110**. This notification is delivered via the questionnaire viewer **1216** running on the second user device **1006b**. The user must then approve that they are the one using the first user device **1006**. If the user does not approve this within a specific time window, for example within 60 seconds, the process must be restarted.

Another authentication input method uses an alphanumeric value comparison method. Here, when the user uses the first user device **1006a**, e.g., at the barrier **1110** or the assigned code (as described below), the main entry system **1002** displays an alphanumeric value to the user, e.g., on the digital display of the QP **1314** located proximate to the harrier **1110**. The user must enter his value into the questionnaire viewer **1216**, which is then sent to the authorization application **1412** for comparison. If the user-provided input matches the value displayed to him/her, this authorization check is complete.

A further authentication input method uses proximity of the second user device to the radio module on the main entry system **1002**. Here, when the user uses the first user device **1006a** or the assigned code (as described below), e.g., at the barrier **1110**, the radio module on the QP proximate to the barrier **1110** is configured to broadcast. If the second user device **1006b** is within proximity of the radio, the questionnaire viewer **1216** running on the second user device **1006b** communicates to the authorization system **1004** that this user is in close proximity to the entry system **1002**, which may be used as an authentication input.

Yet another input method uses biometric data, such as facial recognition and/or voice recognition. Here, when the user uses the first user device **1006a** or the assigned code (as described below), e.g., at the barrier **1110**, the main entry system **1002** requests a facial and/or voice recording from the user, preferably via the questionnaire viewer **1216** running on the second user device **1006b**. The questionnaire viewer **1216** may capture the user's facial image by the camera **1208** on the second user device **1006b**. Alternatively, the authentication input on the main entry system **1002** can be used for image capture. The camera may be mounted near the barrier **1110**. For voice recognition, the user's voice may be recorded by the questionnaire viewer **1216** using the microphone **1210** on the second user device **1006b**. Alternatively, the authentication input device **1312** on the main entry system **1002** may be used for voice capture. The authentication input device **1312** may be equipped with a microphone that may be mounted near the barrier **1110**.

Each user may use the first user device **1006a** as the first factor for identity input. The first user device is assigned a unique identifier which is transmitted to the main entry system **1002** when the user uses the first user device **1006a** to request access. Alternatively, the user may be assigned a

code to be entered into the main entry system **1002** to request access. The assigned code may be entered, e.g., on a keypad provided by the identity input device **1310** on the main entry system **1002**. The unique identifier is the user's unique value that is used for identity input verification. Depending on how the user's input methods are configured, it is either the unique identifier provided on the first user device **1006a** or the manual code that the user enters.

The administration device **1104** is used by administrators for configuring and monitoring the main entry system **1002** and/or the authorization system **1004**. The administration device **1104** may be a computer, smart phone, or tablet connected wirelessly or by wire to the network **1100** via a network interface **1500**, which is similar to the previously described network interfaces **1300**, **1400**. The administrative device **1104** also preferably includes one or more of the following modules: a user interface **1502**, a database **1504**, and a notification handler **1506**. The user interface **1502** is used by the administrator to access and configure the main entry system **1002** or the authorization system **1004**. The database **1504** is used to store system configurations and load user data. The notification handler **1506** is used to receive notifications of abnormal events observed by the authorization system. An example of an event would be when a user has failed to the authorization checks multiple times.

The system **1000** may also include a remote questionnaire pointer (RQP) **1102** which is preferably a permanently fixed device, such as a sign, that may be connected to the network **1100** (hence the broken line connecting the RQP **1102** with the network **1100**). Examples of the QP **1102** may be a parking sign, a sign along a walkway, or a sign at the barrier **1110** that provides instructions to the user seeking to enter a building/parking lot. The RQP **1102** preferably provides a URL link to a questionnaire/survey that visitors of a building must fill out before entry. The sign may be an electronic sign that is connected to the network **1100**. The QP **1102** provides the URL link by giving a static QR Code or phone number for receiving the URL via text. More advanced implementations may broadcast the URL via a radio embedded into the device. In the latter, the sign may be equipped with a solar panel to power the radio. The RQP **1102** may be controlled by the access controller similarly to the one QP **1314** described above for questionnaire delivery.

FIG. 12 is a flow chart showing a high-level operation of the exemplary entry system **1000**. Once the system is configured by the administrator (box **1600**), the system **1000** may be operated to control access to a secure space **114**. Once the user presents his/her unique identifier (identity input) to the main entry system **1002** (box **1602**), e.g., by scanning his/her first user device **1006a** or entering his/her code into the main entry system **1002**, the main entry system **1002** sends the unique identifier to the authorization system **1004** (box **1604**) for verification by the questionnaire application **1410** (box **1606**). If the identity check fails, entry is denied (box **1610**). If the identity check passes, the questionnaire is sent to the user if one is required (box **1608**). Once the user responds to the questionnaire (box **1612**), the response is sent to the authorization system **1004** for authorization. If the response fails, access is denied. If the response passes, the authorization system **1004** determines whether additional authentication is needed (box **1614**). If not, access is granted (box **1616**). If additional authorization is needed, the user is presented with additional authentication input (box **1618**), e.g., facial and/or voice recognition. Once the user enters the additional authentication input, it is verified by the authorization system **1004** (box **1620**). If

15

verification fails, access is denied (box 1610); if verification passes, access is granted (box 1616).

FIGS. 13 and 14 shows the questionnaire configuration and the authorization configuration, respectively. Referring to FIG. 13, the administrator uses the administrator device 1104 to connect to the authorization system 1004 and the main entry system 1002 via the network 1100 (box 1700). Once connected to the authorization system 1004, the administrator uses the questionnaire application 1410 to set the questions to be presented to the user (box 1702). For each question, the administrator configures it as an authorization check or not (box 1704). The administrator then sets the required answer(s) for each question that is configured as an authorization check (box 1706). The administrator then configures the methods for delivering the questionnaire to the user (box 1708), e.g., by geofencing, SMS, passive pointer, and/or broadcast pointer, and the questionnaire expiration date (box 1710). The administrator then configures the QP 1314 used to display the questionnaire to the user (box 1712). The configuration process may simply a drop down menu with a list of RQPs and main entry systems for the administrator to select and configure. The questionnaire application 1410 then stores the questionnaire and associated configuration data in the database 1402 on the authorization system 1004 (box 1714). The process is then repeated until all required questionnaires are configured.

Referring to FIG. 14, the administrator uses the administrator device 1104 to connect to the authorization system 1004 via the network 1100 (box 1750). The administrator then uses the identity and authentication application 1406 on the authorization system 1004 and the database 1504 on the administrator device 1104 to upload each user's unique identifier, name, and raw authentication input to the database 1402 of the authentication system 1004 (box 1752). The administrator then uses the identity and authentication application 1406 to configure the input methods for entry system 1002 used to capture the user's identity and authentication input (box 1754). The identity and authentication application 1406 then uses the generator 1408 to securely store the authentication input as a hash value (box 1756). The administrator then uses the authorization application 1412 to set the authorization checks required for each particular user or user group to gain access (box 1758). The authorization application 1412 next stores a user data object for each user in the database 1402 of the authentication system 1004 (box 1760) which includes the user's unique identifier, name, authentication input hash, input methods, and associated authorization checks. The authorization application 1412 may then upload the database of user data objects to the main entry system 1002 via the network 1100. Uploading to the user data objects to the main entry system 1002, however, is an alternative approach. Ordinarily, the user data objects are stored only in the authentication system 1004.

FIGS. 15-18 show methods for the exemplary system 1000 to deliver the questionnaire to the user, including via geofencing, passive pointer, radio broadcast pointer, and SMS, respectively. Referring to FIG. 15, which is a flow chart showing a method for geofencing delivery of the questionnaire, the administrator enables the geofencing delivery method (box 1800) and the geofencing perimeter (box 1802) using the questionnaire application 1410. To use geofencing delivery, the user must enable the questionnaire viewer 1216, the location service 1206, and notifications 1210 on the second user device 1006b (box 1804). The user must also login to the questionnaire viewer 1216 (box 1806) and have the login information validated by the questionnaire application 1410 in the authorization system 1004 (box

16

1808). Once logged in, the questionnaire viewer 1216 uses the network interface 1200 on the second user device 1006b to retrieve the questionnaire from the questionnaire application 1410 (box 1812). The questionnaire viewer 1216 configures the OS location service 1206 on the second user device 1006b to provide a push notification when the second user device 1006b is within the geofencing perimeter (box 1814). This configuration is preferably accomplished at the time the questionnaire viewer 1216 is installed on the second user device. Once within the geofencing perimeter, the questionnaire viewer 1216 informs the user that he/she is within the perimeter and checks to determine whether the user needs to fill out a questionnaire by checking the expiration date and completion status of the questionnaire (box 1818). If there is no expired or incomplete questionnaire, no action is taken (box 1816). Otherwise, the questionnaire viewer 1216 uses the OS notification handler to notify the user that a questionnaire is required to be successfully completed (box 1820). Once, the user presses the notification (box 1822), the questionnaire viewer 1216 presents the user with the required questionnaire(s) (box 1824). Once the user completely responds to the questionnaire, the questionnaire viewer 1216 sends the completed response to the questionnaire application 1410 to be processed and stored (box 1826).

Referring to FIGS. 16a and 16b, which show a method for delivery of the questionnaire by passive pointer, the administrator, using the questionnaire application 1410, enables the passive delivery method (box 1830) and configures the particular main entry system 1002 and RQP 1102 to be associated with the questionnaire (box 1832). Once enabled and configured, the main entry system 1002 and the RQP 1102 enables digital displays of the passive pointer (box 1834). The authorization system 1004 then provides the main entry system 1002 and the RQP 1102 with content to display on the digital display (box 1836). The main entry system 1002 and the RQP 1102 display the passive questionnaire pointer on their digital display (box 1838). The user then reads the content shown on the digital display of the main entry system 1002 or the RQP 1102 informing him/her a questionnaire is required (box 1840). Using the second user device 1006b, the user can choose to complete the questionnaire (box 1842) by manually entering the URL or by scanning the QR code on the digital display. To manually enter the URL, the user may use the browser 1212 on the second user device 1006b to enter the URL presented on the main entry system 1002 or the RQP 1102 (box 1844). To scan the QR code, the user uses the camera 1208 on the second user device 1006b to scan the QR code presented by the main entry system 1002 or the RQP 1102 (box 1846) which provides the URL present in the QR code to the browser 1212 on the second user device 1006b (box 1848). In either case, the browser 1212 opens the URL which is lead to the questionnaire application 1410 running on the authorization system 1004 (box 1850). Using the data encoded in the URL, the questionnaire application 1410 presents the user with the require questionnaire(s) on the browser 1212 on the second user device 1006b (box 1852). Once the user responds to the questionnaire(s), the questionnaire application 1410 processes and stores the results in the database 1402 (box 1854).

Referring to FIG. 17, which is a flow chart showing a method for delivery of the questionnaire by broadcast pointers, the administrator, using the questionnaire application 1410, enables the broadcast delivery method (box 1860) and configures the particular main entry system 1002 and RQP 1102 to be associated with the questionnaire (box 1862).

This method uses radio broadcast of the URL as discussed above. Once enabled and configured, the main entry system **1002** and the RQP **1102** enable radio broadcasting of the questionnaire pointer (box **1864**). To use broadcast delivery, the user must enable the questionnaire viewer **1216**, the location service **1206**, and notifications **1210** on the second user device **1006b** (box **1866**). Preferably, the questionnaire viewer **1216** is logged on and is running in the background of the second user device **1006b**. Alternatively, the user may be reminded by the RQP to log in to the questionnaire viewer **1216** on the way to the secure space **114**. The user must also login to the questionnaire viewer **1216** (box **1868**) and have the login information validated by the questionnaire application **1410** in the authorization system **1004** (box **1870**). Once logged in, the questionnaire viewer **1216** configures the OS location service **1206** on the second user device **1006b** to periodically scan for the radio broadcast from the main entry system **1002** or the RQP **1102** (box **1872**). When the broadcast is detected, the questionnaire viewer **1216** uses the network interface **1200** on the second user device **1006b** to provide the questionnaire application **1410** on the authorization system **1004** with the unique identifier of the questionnaire pointer and user information (box **1874**). If there is no expired or incomplete questionnaire, no action is taken (box **1875**). Otherwise, the questionnaire application **1410** sends the required questionnaire(s) to the questionnaire viewer **1216** running on the second user device **1006b** (box **1876**). Once the questionnaire(s) is received, the questionnaire viewer **1216** uses the OS notification handler **1204** to notify the user (box **1878**). Once, the user presses the notification (box **1880**), the questionnaire viewer **1216** presents the user with the required questionnaire(s) (box **1882**). Once the user completely responds to the questionnaire, the questionnaire viewer **1216** sends the completed response to the questionnaire application **1410** to be processed and stored (box **1884**).

Referring to FIGS. **18a** and **18b**, which are a flow charts showing a method for delivery of the questionnaire by text messaging (a.k.a. SMS). The administrator, using the questionnaire application **1410**, enables the SMS delivery method (box **2000**) and configures the particular main entry system **1002** and RQP **1102** to be associated with the questionnaire (box **2002**). Once enabled and configured, the QP **1314** of the main entry system **1002** and/or the RQP **1102** enable their digital displays for the SMS request (box **2004**). The authorization system **1004** then provides the RQP **1102** or QP **1314** with content to display on the digital display (box **2006**). The RQP **1102** or QP **1314** then display the SMS request on their digital displays (box **2008**). The user then reads the content shown on the digital display of the QP of the or the RQP **1102** informing him/her that a questionnaire is required (box **2010**). Using the second user device **1006b**, the user can choose to complete the questionnaire (box **2012**) by requesting the questionnaire by following the instructions on the QP **1314** or the RQP **1102** (box **2012**) and texting the location ID of the RQP or QP to the number specified by the instructions (box **2014**). The messaging service **1404** of the authorization system **1004** receives the text message and parses its contents for the location ID of RQP or QP and sends the location ID to the questionnaire application **1410** (box **2016**). The questionnaire application **1410** determines, by looking to the database **1402**, whether that location ID is valid and whether any questionnaire(s) is associated with that location (box **2017**). If the user provided location ID matches and one or more questionnaires are required, then the location ID is valid, and the messaging service **1404** replies back to the text message with an URL

to access the questionnaire and instructions for filling out the questionnaire (boxes **2018** and **2020**). The user can press on the URL link in the text message or copy and paste the URL in the browser **1212** on the second user device **1006b** (box **2021**) to access the questionnaire. The browser **1212** then opens the URL which is lead to the questionnaire application **1410** running on the authorization system **1004** (box **2022**). Using the data encoded in the URL, the questionnaire application **1410** presents the user with the require questionnaire(s) on the browser **1212** of the second user device **1006b** (box **2024**). Once the user responds to the questionnaire(s), the questionnaire application **1410** processes and stores the results in the database **1402** (box **2026**).

FIG. **19** is a flow chart showing the identity input process the user must follow to gain access to the secure space **114**. The identity input process is generally discussed above and shown in box **1602** of FIG. **12**. The administrator uses the authorization application **1412** (accessed via the administrator device **1104**) to select the kind of identity input to be used by the user (box **2100**). Two kinds of identity inputs may be selected: manual code entry (box **2116**) or first user device (box **2102**). The administrator selects one or both input methods for an individual user. The administrator may also select no identity for a particular user, in which case, no identity input is needed when that user requests entry to the secure space **114** (other requirements, such as questionnaire and/or facial or voice recognition may still be applicable to that user to gain entry).

For manual code input, the authorization application **1412** requests a new manual entry code for the user from the identity and authentication application **1406** (box **2118**), causing the identity and authentication application **1406** to generate a unique manual entry code (unique identifier) and to provide it to the authorization application **1412** (box **2120**). The authorization application **1412** stores the input method and entry code (unique identifier) in the user data object for that particular user (box **2122**), preferably in the database **1402** of the authorization system **1004**. The authorization application **1412** then displays the manual entry code to the administrator (box **2124**) who then provides it to the user (box **2126**). The user then manually enters the code into the identity input device **1310** of main entry system **1002** (box **2126**).

For using the first user device **1006a**, the authorization application **1412** requests a new first user device **1006a** unique identifier for the user from the identity and authentication application **1406** (box **2104**), causing the identity and authentication application **1406** to obtain the serial number and unique identifier of the first user device **1006a** from the database **1402** and to provide them to the authorization application **1412** (box **2106**). The authorization application **1412** stores the input method and unique identifier in the user data object for that particular user (box **2108**). The authorization application **1412** then displays the serial number of the first user device **1006a** to the administrator (box **2110**) who then provides the first user device to the user for use when requesting entry to the secure space **114** (box **2112**). To initiate entry request, the user scans the first user device **1006a** at the identity input device **1310** of main entry system **1002** (box **2114**).

In either event (manual code entry or use of the first user device **1006a**), the identity input device **1310** sends the received identity input to the access controller **1306** on the main entry system (box **2130**). The access controller **1306** then forwards the identity input to the authorization application **1412** running on the authorization system **1004** to initiate authentication check (box **2132**).

19

Once the identity input is completed, the system **1000**, if configured to do so for the particular user, conducts an authentication check to verify that the user who presented the identity input is allowed to gain entry to the secure space **114** (authentication). FIGS. **20-23** show different authentication checks that may be selected by the administrator for a particular user. The administrator may select approval from 1) the second user device **1006b** (FIG. **20**); 2) alphanumeric input (FIG. **21**); 3) proximity to the main entry system (FIG. **22**); and 4) biometric data (FIG. **23**).

Referring to FIG. **20**, which is a flow chart showing the authentication input using the second user device **1006b** (box **2200**), the access controller **1306** disables the authentication input device **1312** on the main entry system **1002** (box **2202**). To initiate authentication, the user logs into the questionnaire viewer **1216** and enables notifications on the second user device **1006b** (box **2104**). Preferably, the questionnaire viewer **1216** is logged on and is running in the background of the second user device **1006b**. Alternatively, the user may be reminded by the RQP to log in to the questionnaire viewer **1216** on the way to the secure space **114**. After receiving the identity input from the access controller **1306** (box **2132** in FIG. **19**), the authorization system **1004** sends a notification to the second user device **1006b** (box **2206**). Through the OS notification handler **1204** and the questionnaire viewer **1216** on the second user device **1006b**, the user is presented with a notification and instructions that additional authentication is needed (box **2208**). When the user presses the notification, the questionnaire viewer **1216** is launched to present the user with instruction to provide approval for use of identity at the particular main entry system **1002** (box **2010**). The user either approves (box **2014**) the use or not (box **2012**) which is sent to the authorization application **1412** on the authorization system **1004**. In either case, the authorization application **1412** uses identity input from the access controller **1306** and authentication input from the questionnaire viewer **1216** to determine whether access should be granted (box **2016**).

Referring to FIG. **21**, which is a flow chart showing the authentication input using alphanumeric value (box **2300**), the access controller **1306** disables the authentication input device **1312** (box **2302**), and enables the digital display of the QP **1314** on the main entry system **1002** (box **2304**). The user is shown an alpha numeric value on the digital display of the QP **1314** (box **2304**). The user then logs into the questionnaire viewer **1216** and enables notifications on the second user device **1006b** (box **2306**). Preferably, the questionnaire viewer **1216** is logged on and is running in the background of the second user device **1006b**. Alternatively, the user may be reminded by the RQP to log in to the questionnaire viewer **1216** on the way to the secure space **114**. Once login is verified and after the authorization system **1004** receives the identity input from the access controller **1306** (box **2132** in FIG. **19**), the authorization system **1004** sends an alphanumeric code to the access controller **1306** on the main entry system **1002** to be displayed to the user on the digital display of the QP **1314** for the user (box **2308**), and sends a notification to the second user device **1006b** (box **2310**). Through the OS notification handler **1204** and the questionnaire viewer **1216** on the second user device **1006b**, the user is presented with a notification and instructions that additional authentication is needed (box **2312**). When the user presses the notification, the questionnaire viewer **1216** is launched to present the user with instruction to enter the alphanumeric value provided on the digital display of the QP **1314** (box **2314**). The user then enters the alphanumeric value using the questionnaire viewer **1216** on the second

20

user device **1006b** (box **2316**). The questionnaire viewer **1216** then sends the alphanumeric value entered by the user to the authorization application **1412** on the authorization system **1004** (box **2318**) which uses the identity input from the access controller and the alpha numeric value input to determine whether access should be granted (box **2320**).

Referring to FIG. **22**, which is a flow chart showing the authentication check using the proximity of the second user device **1006b** to the main entry system **1002** (box **2300**), the access controller **1306** disables the authentication input device **1312** (box **2302**), and enables the radio on the QP **1314** of the main entry system **1002** for broadcasting (box **2304**). The user then logs into the questionnaire viewer **1216** and enables notifications and location service on the second user device **1006b** (box **2406**). Preferably, the questionnaire viewer **1216** is logged on and is running in the background of the second user device **1006b**. Alternatively, the user may be reminded by the RQP to log in to the questionnaire viewer **1216** on the way to the secure space **114**. Once login is verified and after the authorization system **1004** receives the identity input from the access controller **1306** (box **2132** in FIG. **19**), the questionnaire viewer **1216** configures the OS location service **1206** of the second user device **1006b** to periodically scan for broadcasts from the QP **1314** (box **2408**). Once a broadcast is detected, the questionnaire viewer **1216** uses the network interface **1200** of the second user device **1006b** to provide the authorization application **1412** on the authorization system **1004** with the unique identifier broadcasted by the QP **1314** (box **2410**). That unique identifier is used as the authentication input. The authorization application **1412** then uses the identity input from the access controller **1306** and the authentication input from the questionnaire viewer **1216** to determine whether access should be granted (box **2312**).

Referring to FIG. **23**, which is a flow chart showing the authentication check using biometric data (box **2300**), the access controller **1306** disables the authentication input device **1312** (box **2302**), and enables the digital display of the QP **1314** on the main entry system **1002** (box **2304**). To obtain biometric input from the user, the administrator may use the authentication input device **1312** of main entry system **1002** or the second user device **1006b**. FIG. **24** is a flow chart showing the authentication check using biometric authentication input from the user device **1006b**; and FIG. **25** is a flow chart showing the authentication check using biometric authentication input from the authentication input device **1312** of the main entry system **1002**.

Referring to FIG. **24**, if the administrator elects to use the second user device **1006b** to obtain biometric data, the access controller **1306** disables the authentication input device **1312** (box **2502**). The user then logs into the questionnaire viewer **1216** and enables notifications and camera and/or microphone access on the second user device **1006b** (box **2504**). Preferably, the questionnaire viewer **1216** is logged on and is running in the background of the second user device **1006b**. Alternatively, the user may be reminded by the RQP to log in to the questionnaire viewer **1216** on the way to the secure space **114**. Once login is verified and after the authorization system **1004** receives the identity input from the access controller **1306** (box **2132** in FIG. **19**), the authorization system **1004** sends a notification to the second user device **1006b** (box **2506**). Through the OS notification handler **1204** and the questionnaire viewer **1216** on the second user device **1006b**, the user is presented with a notification and instructions that additional authentication is needed (box **2508**). Here, the administration can select facial recognition (box **2510**) or voice recognition (box **2516**) as

21

the biometric data. If facial recognition is selected, when the user presses the notification, the questionnaire viewer **1216** is launched to present the user with instructions to provide facial recognition using the camera **1208** on the second user device **1006b** (box **2512**). The questionnaire viewer **1216** also enables the camera **1208** (box **2514**) to take an image of the user's face. If voice recognition is selected by the administrator, when the user presses the notification, the questionnaire viewer **1216** is launched to present the user with instructions to provide voice recognition using the microphone **1210** on the second user device **1006b** (box **2518**). The questionnaire viewer **1216** also enables the microphone **1210** (box **2520**) to record the speaking voice of the user. Once the image or voice is recorded, the questionnaire viewer **1216** uses a hash generator on the second user device **1006b** and the raw authentication input (image or voice recording) to generate an authentication hash value of the authentication input by sending the raw authentication input to the hash generator (box **2522**). The hash generator on the second user device **1006b** uses the raw authentication input to generate the authentication hash value (box **2524**). The questionnaire viewer **1216** then sends the authorization hash value to the authorization application **1412** on the authorization system **1004** to be used as the authentication input (box **2526**). The authorization application **1412** then uses the identity input from the access controller **1306** and the hashed authentication input from the questionnaire viewer **1216** to determine whether access should be granted (box **2528**).

Referring to FIG. **25**, if the administrator elects to use the authentication input device **1312** of the main entry system **1002** to obtain biometric data, the access controller **1306** enables the authentication input device **1312** (box **2530**). Here, the administrator can select facial recognition (box **2532**) or voice recognition (box **2536**) as the biometric data. If facial recognition is selected, after the authorization system **1004** receives the identity input from the access controller **1306** (box **2132** in FIG. **19**), the access controller **1306** enables the camera on the authentication input device **1312** on the main entry system **1002** (box **2534**) to take an image of the user's face. If voice recognition is selected, after the authorization system **1004** receives the identity input from the access controller **1306** (box **2132** in FIG. **19**), the access controller **1306** enables the microphone on the authentication input device **1312** on the main entry system **1002** (box **2538**) to record the talking voice of the user. Once the image or voice is recorded, the authentication input device **1312** send the raw authentication input (image or voice recording) to the access controller **1306** (box **2540**) which, in turn, sends the raw authentication input to the hash generator **1308** to generate a hash value of the raw authentication input (box **2542**). The access controller **1306** then sends the hash value to the authorization application **1412** on the authorization system **1004** to be used as the authentication input (box **2544**). The authorization application **1412** then uses the identity input from the access controller **1306** and the hashed authentication input to determine whether access should be granted.

FIG. **26** is a flow diagram showing the authorization check process after identity and authentication inputs have been obtained by the authorization application **1412** on the authorization system **1004**. Once the identity input is received, it is checked against the user data object for the particular user in the database **1402** (box **2600**). If the identity input does not match, access is denied. The administrator then uses the authorization application **1412** to select the types of authorization check required (box **2602**). If

22

authentication is enabled, the authorization application **1412** receives the authentication input from the main entry system **1002** or the questionnaire viewer **1216** of the second user device **1006b** (see boxes **2528** of FIG. **24**, **2412** of FIG. **22**, **2320** of FIG. **21**, and **2216** of FIG. **20**) and compares it to the user data object for the particular user in the database **1402** (box **2604**). If the authentication input does not match access is denied. The administrator then uses the authorization application **1412** to select the types of authorization check required (box **2606**). If questionnaire authorization check is enabled, the authorization system **1004** checks whether the authorization application **1412** receives valid response(s) to the questionnaire (box **2608**). If the response(s) is not valid, access is denied. If the response(s) is valid, the authorization application **1412** sends an access granted message to the access controller **1306** on the main entry system **1002** (box **2610**). The access controller **1306**, in turn, release the lock on the barrier to allow the user to enter the secure space **114** (box **2612**).

Although certain presently preferred embodiments of the invention have been specifically described herein, it will be apparent to those skilled in the art to which the invention pertains that variations and modifications of the various embodiments shown and described herein may be made without departing from the spirit and scope of the invention. Accordingly, it is intended that the invention be limited only to the extent required by the appended claims and the applicable rules of law.

What is claimed is:

1. A method for entering a secure space, comprising:

providing a controlled entry system at an entrance of a building, the entry system configured to interact with a plurality of first user devices and a plurality of second user devices, the entry system comprising a main entry system controlling a barrier and an authorization system having a database and configured to validate the first and second user devices and to send a questionnaire, survey or form to the second user device, wherein the second user devices, the main entry system, and the authorization system communicate with each other over a computer network;

reading one of the plurality of first user devices to obtain a unique identification;

determining from the database whether a questionnaire, survey or form is required from information associated with the unique identification stored in the authorization system;

sending an authentication request to one of the plurality of second user devices;

receiving an authentication input in response to the authentication request from the one of the plurality of second user devices;

if a questionnaire, survey or form is required, sending such questionnaire, survey or form to and receiving a response to the questionnaire, survey or form from the one of the plurality of second user devices;

comparing the unique identification, the authentication input, and the response with information from a database; and

granting access to the secure space only if the unique identification, the authentication input, and the response match the information of the database.

2. The method of claim 1, wherein the one of the plurality of first user devices comprises an identification card, an employee badge, a bar code, a QR code, or combinations thereof.

23

3. The method of claim 1, wherein the one of the plurality of second user devices comprises a smart phone, a smart watch, a laptop, or a tablet.

4. The method of claim 1, wherein the authentication input comprises a picture of a user's face, a voice recording of the user's voice, an assigned code, or proximity of the second user device to the entrance.

5. The method of claim 1, wherein sending a questionnaire, survey or form occurs when the one of the plurality of second user devices is within a predetermined distance of the entrance.

6. The method of claim 1, wherein reading one of the plurality of first user devices occurs at an identity input device.

7. The method of claim 6, wherein the identity input device comprises one of a RFID reader, a bar code reader, a QR reader, a cell phone, a tablet computer, or combinations thereof.

8. The method of claim 1, wherein reading one of the plurality of first user devices occurs when the one of the plurality of first users is within the predetermined distance of the entrance.

9. A controlled entrance, comprising:

an entryway in a building, the entryway including a barrier operable to permit access to the building through the entryway;

an identity input device for determining the proximity of an individual seeking access to the building through the entryway, the identity input device configured to read a first user device associated with a unique identification for the individual;

an authentication input device configured to receive an authentication input from the individual;

an authorization system operably associated with the entryway and configured to determine whether the individual needs to submit a questionnaire, survey or form, to provide a pointer or link thereto and to receive the questionnaire, survey or form from a second user device associated with the individual; and

an access controller operably associated with the barrier and the authorization system for operating the barrier upon determination that the unique identification, the authentication input, and the completed questionnaire, survey or form match the information from a database associated with the authorization system.

10. The controlled entrance of claim 9, wherein the identity input device comprises any of a RFID reader, a

24

fingerprint scanner, a retinal scanner, a camera, a keypad, a writing pad, a voice recorder, a bar code reader, a QR reader, a cell phone, a tablet computer, or combinations of them.

11. The controlled entrance of claim 10, wherein the identity input device is configured for multi-factor authentication.

12. The controlled entrance of claim 10, wherein the identity input device is configured to identify the individual when the individual is within a defined area adjacent to the entrance.

13. The controlled entrance of claim 10, wherein a temperature sensor and/or camera is operably associated with the entryway for determining at least one of the body temperature and facial expression of the individual.

14. The controlled entrance of claim 10, wherein a wired or wireless interface interconnects the identity input device with the control system and with the authorization system.

15. The controlled entrance of claim 14, wherein the wireless interface operate via at least one of cellular (3G/4G/5G), Bluetooth, Bluetooth Lower Energy, WiFi, TCP/IP, near field communication (NFC).

16. The controlled entrance of claim 9, wherein the authentication input device comprises any of a camera, a microphone, or combinations of any of them.

17. The controlled entrance of claim 9, wherein the questionnaire, survey or form comprises a non-disclosure agreement (NDA), a security clearance, a safety protocol, a code of conduct, a non-compete agreement, an engagement census, a health questionnaire, or combinations thereof.

18. The controlled entrance of claim 9, wherein: the barrier is a door pivotally or movably mounted to a doorframe, wherein the authorization system is configured to control a lock or latch to allow or restrict access through the barrier;

and the identity input device is mounted proximate the doorframe.

19. The entryway of claim 9, wherein a database contains individual identification information, questionnaire information, and authentication information.

20. The entryway of claim 9, wherein the authorization system determines whether the questionnaire, survey or form is required when the one of the plurality of second user devices is within a predetermined distance of the entrance.

* * * * *